

AKADEMIKA

BEL-ADA

**ANALOG TO DIGITAL AND
DIGITAL TO ANALOG CONVERTER
TRAINER**

EXPERIMENTAL MANUAL

.... A MESSAGE FROM



BEL-ADA Analog to Digital/Digital to Analog trainer is the first of its kind using actual techniques of Analog to Digital/Digital to Analog conversion. It is our vision to furnish products, which are reliable and convenient to use. AKADEMIKA offers a high degree of expertise in developing efficient products.

Company's Motto

- Light years ahead – refers to leadership.

We strive to be the best in what we do and maintain high standards in the areas of

• Design • Quality • Value • Delivery • Support

We are indeed light years ahead of the competition in this field. We guarantee our valued customers great satisfaction.

Important Information

As you will move through this manual you will quickly discover that we have complete, truly innovative & superior training products. We are so committed to quality that we back our products with a complete and comprehensive warranty.

AKADEMIKA LAB SOLUTIONS

Sr. No. 15/8/1, Unit No.9, Kruti Industrial Estate, Opp.
Karishma Society, Off Karve Road, Kothrud, Pune,
Maharashtra- 411038
INDIA

Contact No.:+91-7447438443

Email: lab@akademika.in

www.akademika.in

SAFETY RULES

- Read carefully and follow the instructions mentioned in this manual. This user manual includes all the important points about the installation, use and the maintenance of the product. Keep this manual always with you, for quick reference.
- After unpacking the product, arrange all the accessories in proper order, so that their integrity is checked with the packing list. Also, ensure that the accessories have no visible damage.
- Before connecting the power supply to the kit, be sure that the jumpers and the connecting chords are connected correctly, as per the experiment.
- This kit must be employed only for the use for which it has been conceived, i.e. as educational kit and must be used under the direct survey of expert personnel. Any other use is inadvisable and dangerous too. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.
- In case of any fault or malfunctioning in the trainer kit, turn off the power supply. Please do not tamper the kit. If you require our service, kindly contact the service centre for technical assistance.
- BEL-ADA is High Level, High Quality Analog-Digital Trainer kit. It combines all essential functions of Analog and Digital Experiment with removable breadboard, Includes DC power supply, Function Generator, Two Pulse Switches, 8 ohm 0.25W Speaker
- The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

This kit is warranted against defects in workmanship and materials. Any failure due to defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we would charge for repairs, regardless of the warranty period. The warranty does not cover include perishable items like connecting chords, crystals, etc. and other imported items.

Conditions and Limitations

The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product will be Akademika Lab Solutions sole judgment. The defective product will be replaced with a new one or repaired, without charge or with charge.

In the warranty period if the service is needed, the purchaser should get in touch with the service center or the sales outlet. The purchaser should return the product to the service center or the sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of the kit has to be mentioned specifically. We return the product to the purchaser at our expense. In case the warranty does not cover the product on Akademika Lab Solutions judgment, we would repair the product after obtaining prior permission from purchaser who would receive an estimate statement about the repairing charges. In such cases, Akademika Lab Solutions bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but they have no rights to stretch the meaning of original warranty contents or to offer an additional warranty. Akademika Lab Solutions does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solutions is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs, maintenance and up gradation of products, be sure to contact our service center or the sales outlet.

INDEX

Sr. No.	Chapter	Page No.
1	Technical Specifications	5
2	Functional Block	6
3	Introduction	7
4	Circuit Diagram	8
5	Experiment No. 1 Study of D/A conversion using binary weighted Resisters and measure its resolution	13
6	Experiment No.2 Study of D/A conversion using R/2R ladder Network and measure its resolution	19
7	Experiment No.3 Study of D/A conversion using Monolithic IC DAC0800 & to measure its Settling time and effect of settling Time	25
8	Experiment No.4 Study of Working of A/D conversion Using counter ramp method	39
9	Experiment No.5 Study of A/D conversion using monolithic IC and to study full scale adjustment for unipolar And bipolar signals of ADC	45
10	Experiment No.6 Study to measure the conversion time of ADC	49
11	Experiment No. 7 Study to measure their solution of LTC1278 IC	53

TECHNICAL SPECIFICATION

Sine Wave Generator:

- Amplitude : 0 to 5V.
- Frequency : 1 KHz

Variable DC Generator:

- Voltage Range : -12V to +12V

Clock Generator:

- Frequency Range : 0.5Hz, 1Hz, 2Hz, 4Hz

Digital To Analog Converter:

- 8 bit Binary Weighted Resistors DAC
- 8 bit Ladder type D to A Converter.
- 8 bit D to A Converter using monolithic IC.

Analog To Digital Converter:

- 8 bit discrete ADC
- 12 bit successive approximation monolithic ADC IC.

Interconnections:

- This is done with the help of supplied connecting patch chords (2 mm banana socket).

Binary Counter:

- 8 bit binary counter with external clock frequency.

Test Points:

- 5 onboard test points avails the user to observe various signals on Oscilloscope

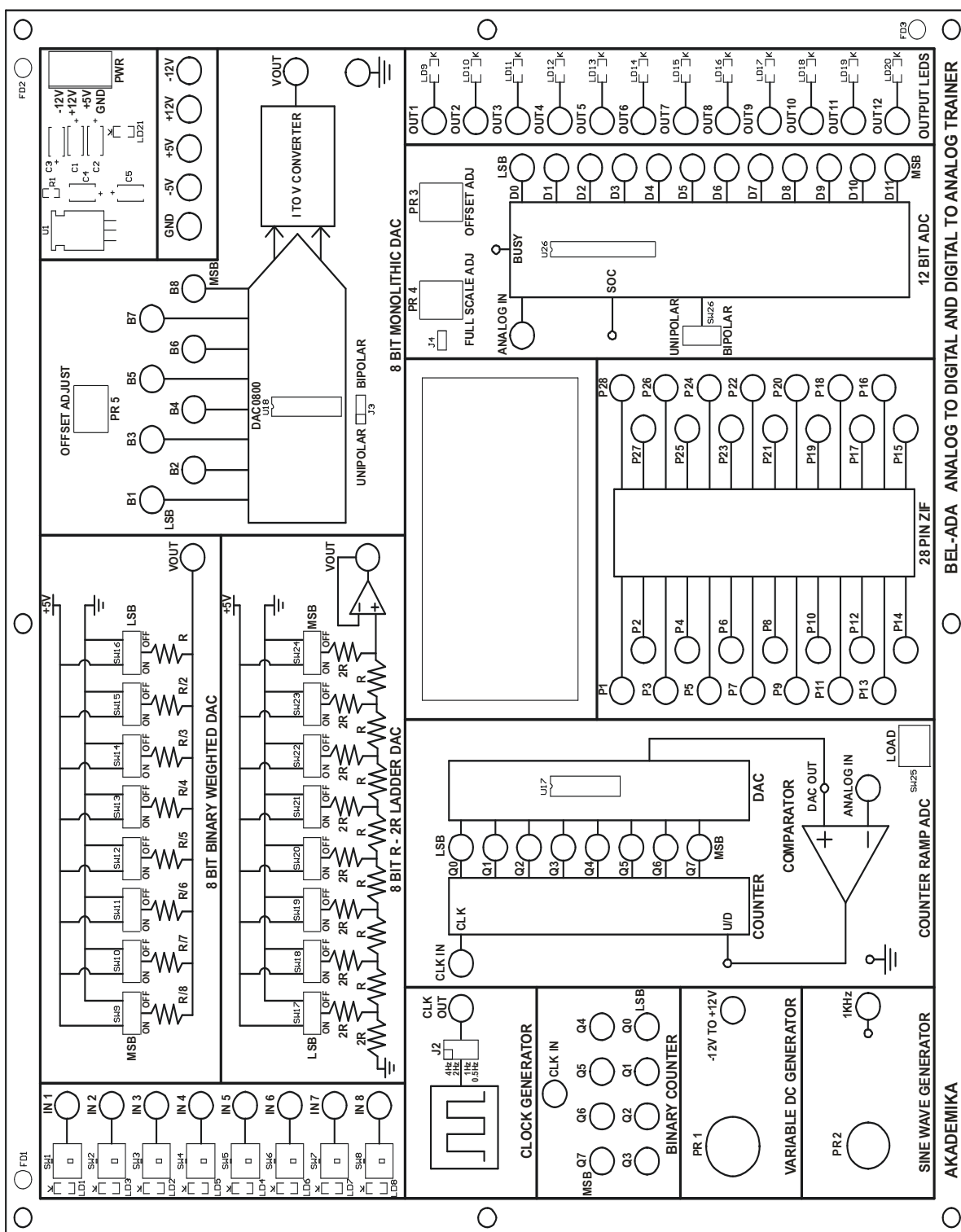
Power Supply:

- -12V, +12V, +5V, GND

Accessories:

BEL-ADA			Quantity
1	Short Links PCS2 Patch cord	Male To Male	15
2	Long Links PCS2 Patch cord	Male To Male	5
3	BEL-ADA EXPT Manual		1
4	Power Supply		1
5	Power cord		1

FUNCTIONAL BLOCK

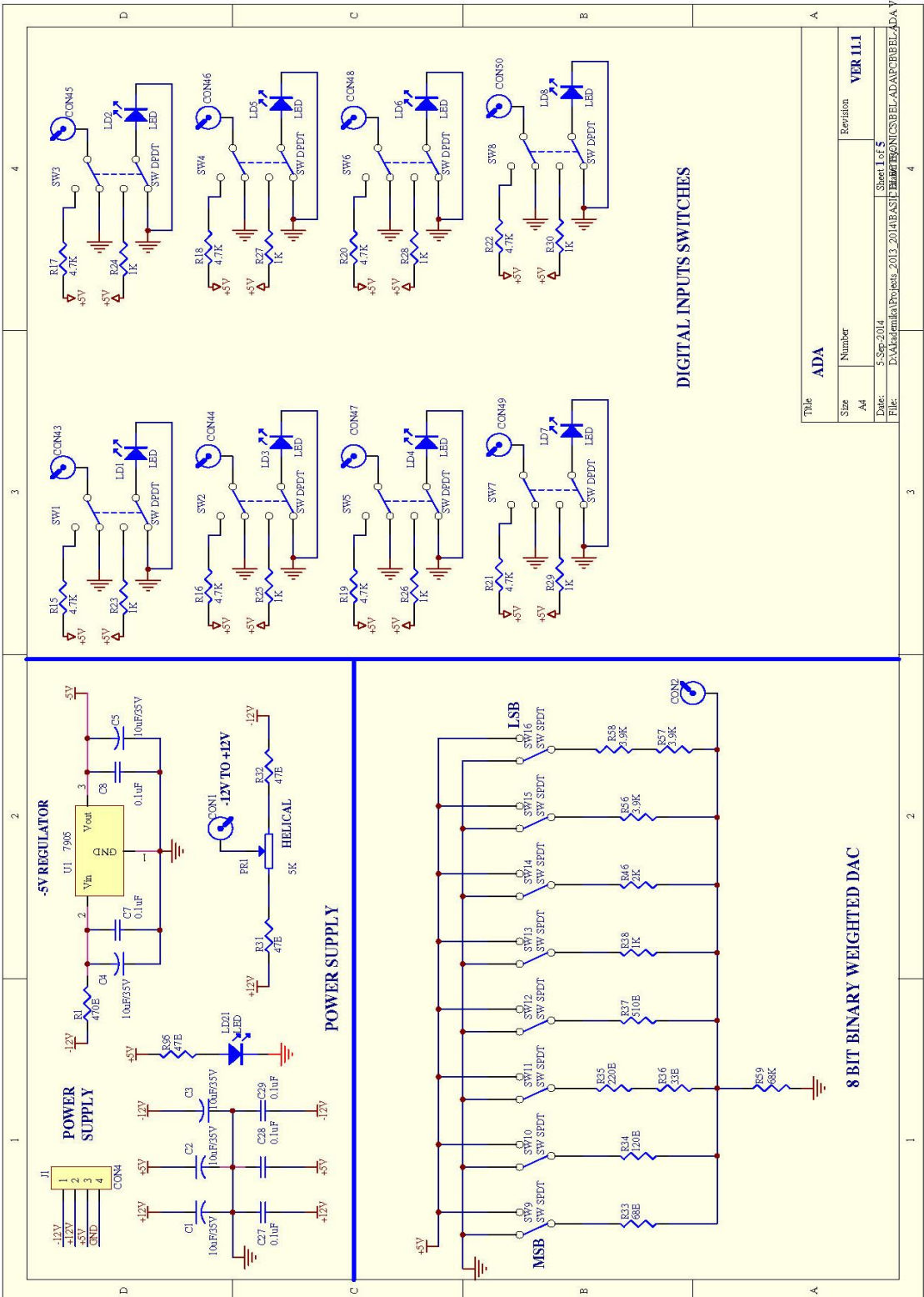


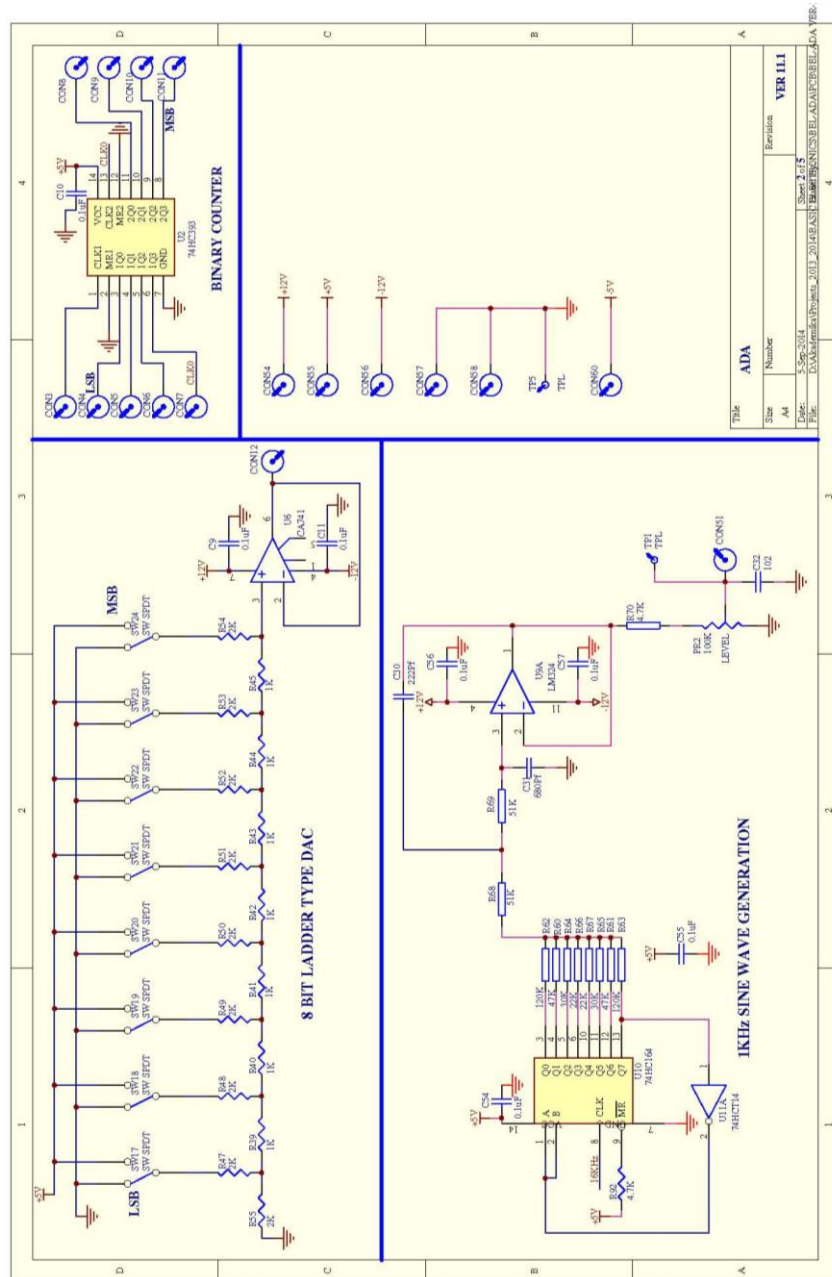
INTRODUCTION

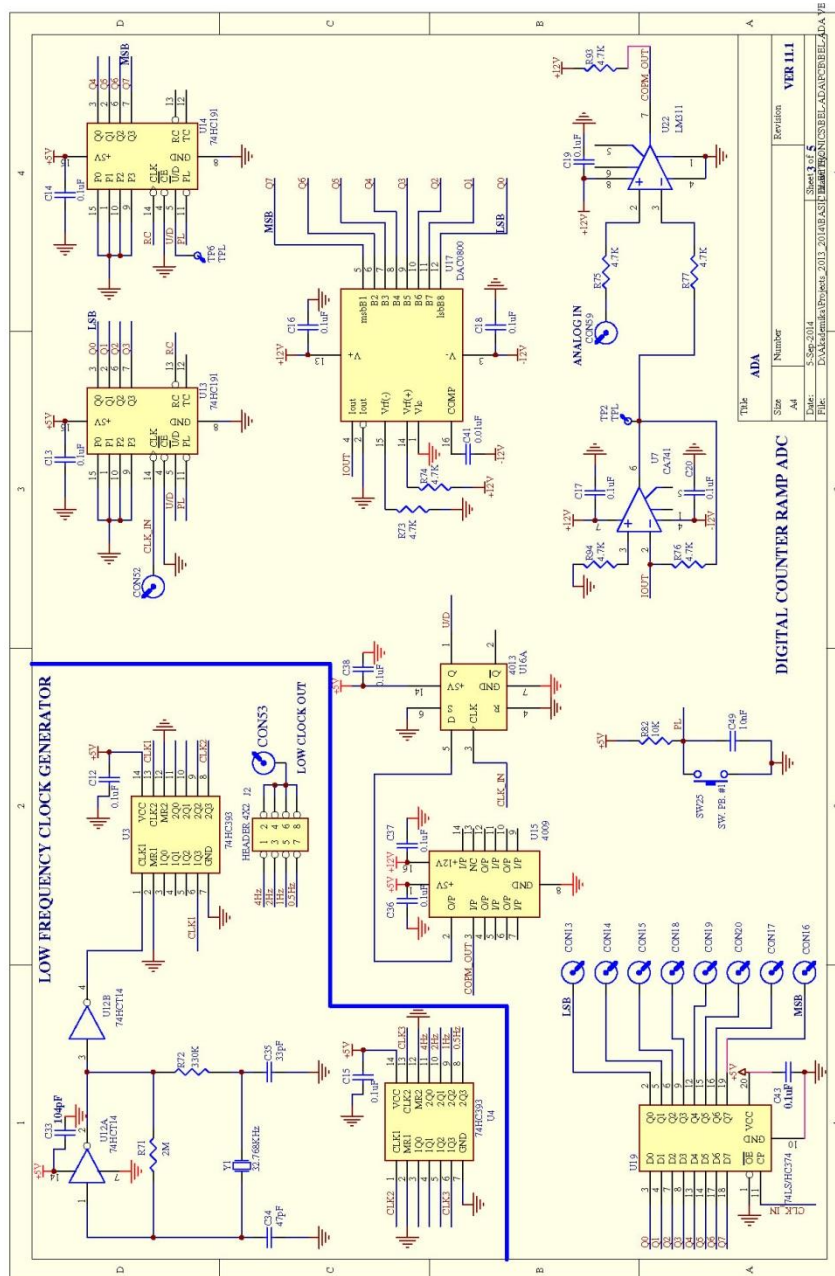
The **Analog to Digital and Digital to Analog Trainer** is a very versatile and descriptive with onboard regulated power supply. The trainer covers topics of Digital to Analog and Analog to Digital Conversion using Different methods. D/A and A/D converters form two very important aspects of digital data processing. D/A conversion involve translation of Digital information to Analog information. As an example the output of Digital System might be changed to analog form for the purpose of driving a pen recorder similarly an analog signal is required for servomotors which drive the cursor arms of a plotter.

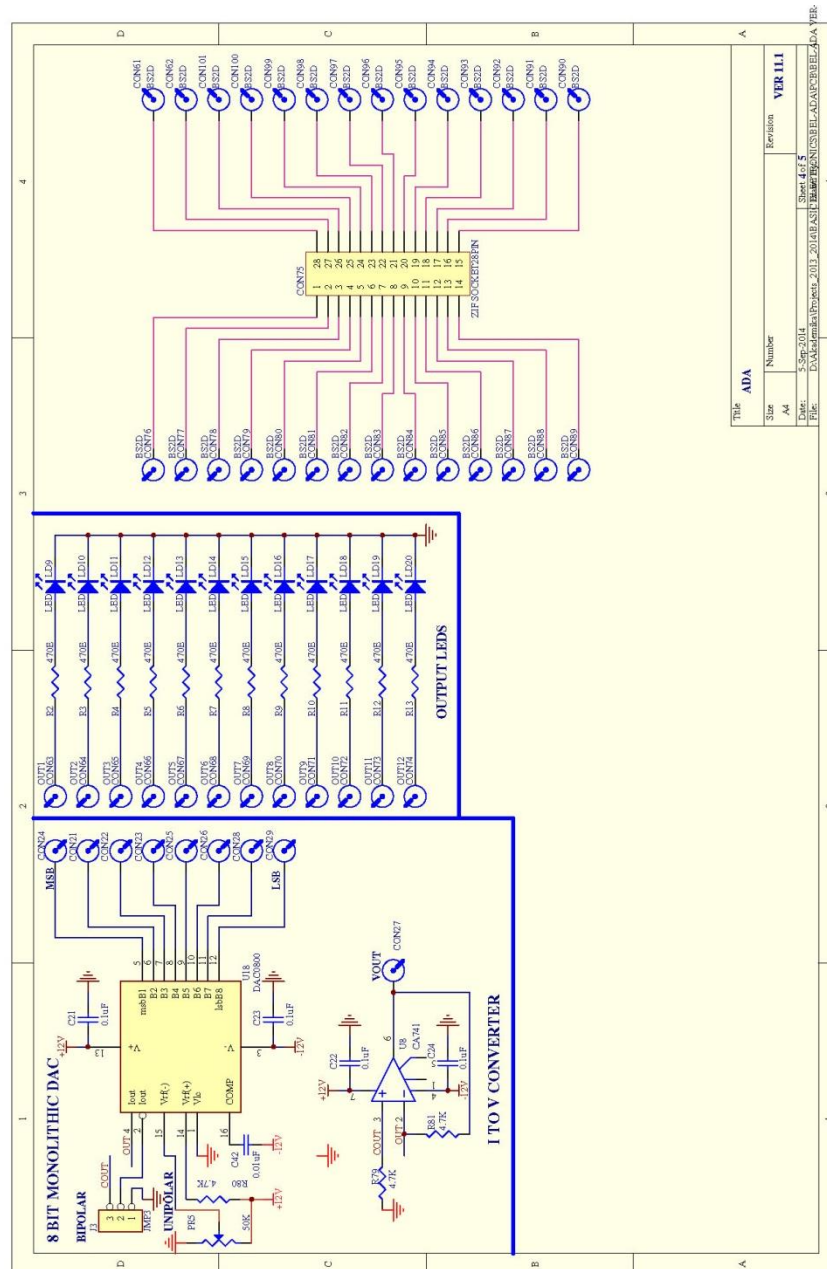
Interconnections between circuits allow easy designing using on board resources. It has 8 slider switches for providing inputs to D/A converters and 12 LEDs to observe the output. It has 8 bit onboard binary counter to see the ramp waveform from 8 BIT MONOLITHIC DAC. Onboard 28 pin ZIF socket and a bread board are provided for developing user defined applications. The system comes with descriptive experimental manual.

CIRCUIT DIAGRAM











EXPERIMENT NO.1

EXPERIMENT NO.1

NAME:-

Study of D/A Conversion Using Binary Weighted Resistor And Measurement Of Its Resolution

OBJECTIVE:-

To study 8 bit Digital to Analog Conversion using Binary Weighted Resistors Method and measures its resolution.

EQUIPMENTS:-

- BEL-ADA Board
- Power supply
- DMM

THEORY:-

Digital to Analog Conversion involve conversion of Digital signal to Analog signal. It is an integral part of any A/D converter. This can be easily accomplished by designing a resistive network that will change each digital level into an equivalent binary weighted voltage. There are two methods by which we can do this process.

First method is using binary weighted resistor method.

In this method the binary coded decimal code is used for the representation of the decimal number. The LSB which is at right most location is expressed as $(1 \text{ or } 0) \cdot 2^0$, the next bit is $1 \text{ or } 0 \cdot 2^1$ and so on, where 0 indicates 0V and 1 indicates 5V). Thus the weight of each column from right to left in n bit converter are $1, 2, 4, 8, \dots, 2^{n-1}$.

Following criteria can be applied to this divider.

- There must be one input resistor for each digital bit.
- Beginning with the LSB, each following resistor value is one half of the size of the previous resistor.
- The full scale output voltage is equal to the positive voltage of the digital I/P signal.
- The LSB has a weight of $1/(2^n - 1)$, where n is the number of input bits.
- The change in output voltage due change in LSB is equal to the $V/(2^n - 1)$, where V is the digital input voltage level.
- The output voltage can be found for any digital input signal by using the following modified form of Mill man's theorem.

$$V_{OUT} = (V_0 \cdot 2^0 + V_1 \cdot 2^1 + V_2 \cdot 2^2 + \dots + V_{N-1} \cdot 2^{N-1}) / (2^N - 1)$$

Where $V_0, V_1, V_2, \dots, V_{N-1}$ are the digital input voltage level (0V or 5V) and n is the number of input bits.

RESOLUTION:-

Resolution indicates the smallest increment of the output corresponding to a 1 LSB input code change. For example for 8 bit DAC, $2^8 = 256$ codes, so the resolution is $1/256$ of the output range.

PROCEDURE:-

- Connect the trainer kit with its power supply.
- Give the desired logic input to the 8 BIT BINARY WEIGHTED DAC from on board slider switches as shown below in FIGURE-1.1
- Measure the equivalent output voltage at V_{OUT} of 8 BIT BINARY WEIGHTED DAC for different digital input voltage level with the help of DMM.
- Observed the calculated and observed readings for different input levels.

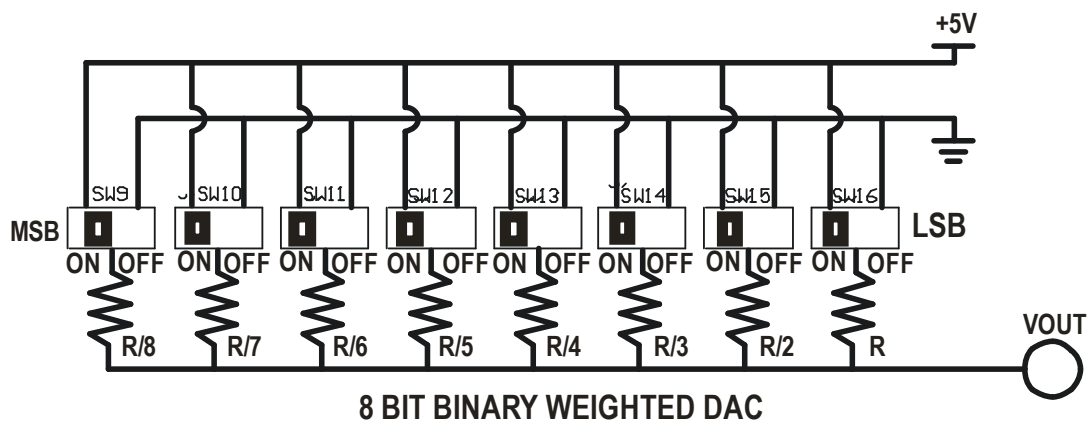


FIGURE-1.1

OBSERVATIONS:-

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Calculated O/P Voltage (in V)	Observed O/P Voltage (in V)
0	0	0	0	0	0	0	0	00.00	0.000
0	0	0	0	0	0	0	1	0.019	0.021
0	0	0	0	0	0	1	0	0.039	0.042
0	0	0	0	0	0	1	1	0.058	0.064
0	0	0	0	0	1	0	0	0.078	0.083
0	0	0	0	0	1	0	1	0.098	0.104
0	0	0	0	0	1	1	0	0.117	0.126
0	0	0	0	0	1	1	1	0.137	0.147
0	0	0	0	1	0	0	0	0.156	0.166
0	0	0	0	1	0	0	1	0.176	0.187
0	0	0	0	1	0	1	0	0.196	0.209
0	0	0	0	1	0	1	1	0.215	0.230
0	0	0	0	1	1	0	0	0.235	0.249
0	0	0	0	1	1	0	1	0.254	0.271
0	0	0	0	1	1	1	0	0.274	0.292
0	0	0	0	1	1	1	1	0.294	0.313
0	0	0	1	0	0	0	0	0.313	0.328
0	0	0	1	0	0	0	1	0.333	0.347
0	0	0	1	0	0	1	0	0.352	0.369
0	0	0	1	0	0	1	1	0.372	0.390
0	0	0	1	0	1	0	0	0.392	0.409
0	0	0	1	0	1	0	1	0.411	0.430
0	0	0	1	0	1	1	0	0.431	0.451
0	0	0	1	0	1	1	1	0.450	0.472
0	0	0	1	1	0	0	0	0.470	0.492
0	0	0	1	1	0	0	1	0.490	0.512
0	0	0	1	1	0	1	0	0.509	0.534
0	0	0	1	1	0	1	1	0.529	0.555
0	0	0	1	1	1	0	0	0.549	0.574

0	0	0	1	1	1	0	1	0.568	0.596
0	0	0	1	1	1	1	0	0.588	0.618
0	0	0	1	1	1	1	1	0.607	0.639
0	0	1	0	0	0	0	0	0.627	0.659
0	0	1	0	0	0	0	1	0.647	0.680
0	0	1	0	0	0	1	0	0.666	0.701
0	0	1	0	0	0	1	1	0.686	0.722
0	0	1	0	0	1	0	0	0.705	0.742
0	0	1	0	0	1	0	1	0.725	0.763
0	0	1	0	0	1	1	0	0.745	0.784
0	0	1	0	0	1	1	1	0.764	0.806
0	0	1	0	1	0	0	0	0.784	0.823
0	0	1	0	1	0	0	1	0.803	0.841
0	0	1	0	1	0	1	0	0.823	0.862
0	0	1	0	1	0	1	1	0.843	0.883
0	0	1	0	1	1	0	0	0.862	0.902
0	0	1	0	1	1	0	1	0.882	0.923
0	0	1	0	1	1	1	0	0.901	0.945
0	0	1	0	1	1	1	1	0.921	0.966
0	0	1	1	0	0	0	0	0.941	0.982
0	0	1	1	0	0	0	1	0.960	1.004
0	0	1	1	0	0	1	0	0.980	1.026
0	0	1	1	0	0	1	1	1.00	1.047
0	0	1	1	0	1	0	0	1.019	1.062
0	0	1	1	0	1	0	1	1.039	1.087
0	0	1	1	0	1	1	0	1.058	1.110
0	0	1	1	0	1	1	1	1.078	1.129
0	0	1	1	1	0	0	0	1.098	1.148
0	0	1	1	1	0	0	1	1.117	1.170
0	0	1	1	1	0	1	0	1.137	1.190
0	0	1	1	1	0	1	1	1.156	1.212
0	0	1	1	1	1	0	0	1.176	1.229
0	0	1	1	1	1	0	1	1.196	1.253
0	0	1	1	1	1	1	0	1.215	1.273

0	0	1	1	1	1	1	1	1.235	1.298
0	1	0	0	0	0	0	0	1.254	1.386
0	1	0	0	0	0	0	1	1.274	1.408
1	1	1	1	1	1	1	1	5.00	5.09

CALCULATIONS:-

$$V_{OUT} = (V_0 2^0 + V_1 2^1 + V_2 2^2 + \dots + V_{N-1} 2^{N-1}) / (2^N - 1)$$

Take any combinations e.g. 00011000

$$V_{OUT} = 0 \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 0 \times 2^5 + 0 \times 2^6 + 0 \times 2^7 = (24 \times 5) / (2^8 - 1) = 0.470V$$

Take another one i.e. 00111111

$$V_{OUT} = 1 \times 2^0 + 1 \times 2^1 + 1 \times 2^2 + 1 \times 2^3 + 1 \times 2^4 + 1 \times 2^5 + 0 \times 2^6 + 0 \times 2^7 = (63 \times 5) / (2^8 - 1) = 1.235V$$

RESOLUTION:-

Resolution of 8 bit converter can be calculated by using formula

$$= (1/2^N) \times V_{ref} = 0.019V$$

$$\text{Calculated Value of Resolution} = (1/2^8) \times 5 = 0.019V$$

For observed value subtract any consecutive observed reading like $0.042 - 0.021 = 0.021V$, this will give observed Value of Resolution.

CONCLUSION:-

The analog output voltage can be found for any digital input signal by using the binary weighted resistor network and its resolution for 8 bits is $0.019V(\text{LSB})$

EXPERIMENT NO.2

EXPERIMENT NO.2

NAME:-

Study of D/A conversion using R/2R ladder type method and measurement of its resolution

OBJECTIVE:-

To study 8 bit Digital to Analog Conversion using R/2R Ladder type Method and measure its resolution.

EQUIPMENTS:-

- BEL-ADA Board
- Power supply and
- DMM

THEORY:-

Binary R/2R Ladder is a resistive network whose output voltage is properly weighted sum of the Digital input.

It is constructed of resistors that have only two values. We can see from the FIGURE-2.1 that total resistance looking from any node back towards the terminating resistor or out towards the digital input is 2R. We can use the resistance characteristics of the ladder to determine the output voltage for the various digital inputs. Ladder is composed of linear resistors, it is a linear network and the principle of superposition can be used. This means that the output voltage due to combination of input digital level can be found by simply taking the sum of the output levels caused by each digital input individually. To keep the ladder in perfect balance and to maintain symmetry, the output of ladder should be terminated in a resistance of 2R. This will result in lowering of output voltage, but if the 2R load is maintained constant, the output voltage will still be a properly weighted sum of binary input bits. If load is varied, the output voltage will not be a properly weighted sum hence load should be kept constant.

$$V_A = V/2 + V/4 + V/8 + \dots + V/2^N$$

Where N is the total number of input bits

$$V_A = (V_0 2^0 + V_1 2^1 + V_2 2^2 + \dots + V_{N-1} 2^{N-1}) / 2^N$$

Where $V_0, V_1, V_2, \dots, V_{N-1}$ are the digital input voltage levels (0V or 5V).

PROCEDURE:-

- 1) Connect the trainer kit with its power supply.
- 2) Give the desired logic input to the converter from on board slider switches of 8 BIT R/2R LADDER DAC SECTION as shown below in FIGURE-2.1
- 3) Measure the equivalent output voltage for different digital input voltage level with the help of DMM at VOUT of 8 BIT R/2R LADDER DAC.
- 4) Observed the calculated and observed readings for different input levels.

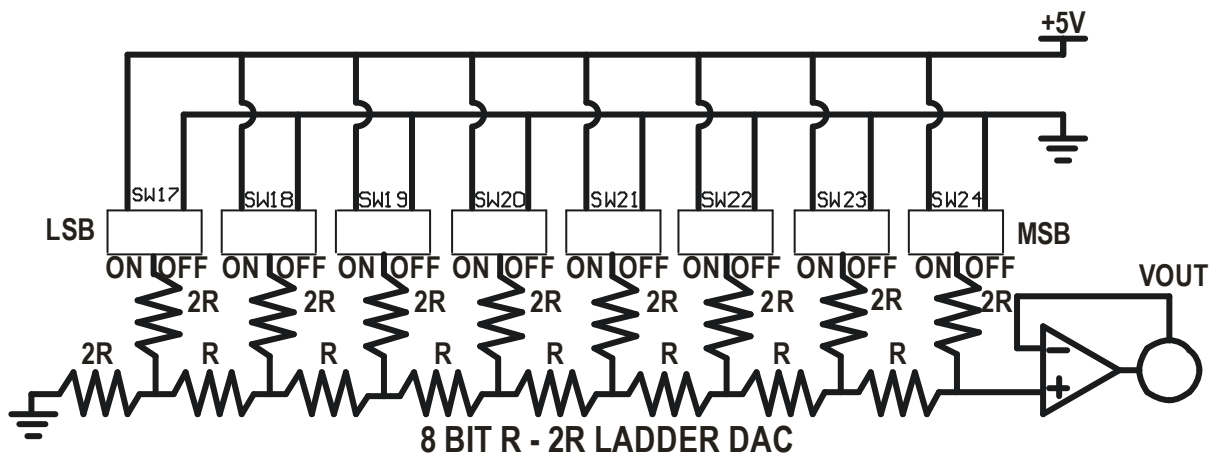


FIGURE-2.1

OBSERVATIONS:-

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Calculated O/P Voltage(in V)	Observed O/P Voltage(in V)
0	0	0	0	0	0	0	0	00.00	0.001
0	0	0	0	0	0	0	1	0.019	0.0186
0	0	0	0	0	0	1	0	0.039	0.0388
0	0	0	0	0	0	1	1	0.058	0.0590
0	0	0	0	0	1	0	0	0.078	0.0788
0	0	0	0	0	1	0	1	0.097	0.0989
0	0	0	0	0	1	1	0	0.117	0.119
0	0	0	0	0	1	1	1	0.136	0.139
0	0	0	0	1	0	0	0	0.156	0.158
0	0	0	0	1	0	0	1	0.175	0.177
0	0	0	0	1	0	1	0	0.195	0.196
0	0	0	0	1	0	1	1	0.214	0.218
0	0	0	0	1	1	0	0	0.234	0.238
0	0	0	0	1	1	0	1	0.253	0.258
0	0	0	0	1	1	1	0	0.273	0.278
0	0	0	0	1	1	1	1	0.292	0.298
0	0	0	1	0	0	0	0	0.312	0.318
0	0	0	1	0	0	0	1	0.332	0.338
0	0	0	1	0	0	1	0	0.351	0.358
0	0	0	1	0	0	1	1	0.371	0.378
0	0	0	1	0	1	0	0	0.390	0.396
0	0	0	1	0	1	0	1	0.410	0.417
0	0	0	1	0	1	1	0	0.429	0.437
0	0	0	1	0	1	1	1	0.449	0.457
0	0	0	1	1	0	0	0	0.468	0.476
0	0	0	1	1	0	0	1	0.488	0.497
0	0	0	1	1	0	1	0	0.507	0.517
0	0	0	1	1	0	1	1	0.527	0.536
0	0	0	1	1	1	0	0	0.546	0.557
0	0	0	1	1	1	0	1	0.566	0.576
1	0	0	0	0	0	0	0	2.499	2.558
1	1	0	0	0	0	0	0	3.749	3.83
1	1	1	0	0	0	0	0	4.374	4.47
1	1	1	1	0	0	0	0	4.687	4.79
1	1	1	1	1	0	0	0	4.843	4.95
1	1	1	1	1	1	0	0	4.921	5.03

1	1	1	1	1	1	1	0	4.960	5.07
1	1	1	1	1	1	1	1	5.000	5.08

CALCULATIONS:-

$$V_A = (V_0 2^0 + V_1 2^1 + V_2 2^2 + \dots + V_{N-1} 2^{N-1}) / 2^N$$

Take any combination for e.g. 1111 1100

$$V_A = (0 \times 2^0 + 0 \times 2^1 + 5 \times 2^2 + 5 \times 2^3 + 5 \times 2^4 + 5 \times 2^5 + 5 \times 2^6 + 5 \times 2^7) / 2^8 = 4.921V$$

Take 1111 0000

$$V_A = (0 \times 2^0 + 0 \times 2^1 + 0 \times 2^2 + 0 \times 2^3 + 5 \times 2^4 + 5 \times 2^5 + 5 \times 2^6 + 5 \times 2^7) / 2^8 = 4.687V$$

RESOLUTION

Resolution of 8 bit converter can be calculated by

$$\text{Formula} = (1/2^N) \times V_{\text{ref}} = 0.019V$$

$$\text{Calculated Value of Resolution} = (1/2^8) \times 5 = 0.019V.$$

For observed value subtract any consecutive observed reading like 0.0388-

$$0.0186 = 0.020V. \text{ This will give observed Value of Resolution.}$$

CONCLUSION:-

The resistance characteristics of the ladder is used to determine the analog output voltage for the various digital inputs and for 8 bits D/A converter, resolution measured is 0.019V.

EXPERIMENT NO.3

EXPERIMENT NO.3

NAME:-

D/A conversion using monolithic IC DAC0800 & to measure its settling time & effect of settling time.

OBJECTIVE:-

To study 8 bit Digital to Analog Conversion using Monolithic IC DAC0800 for various digital data pattern and to measure

- a) Ramp in unipolar and bipolar configuration
- b) Settling time
- c) Effect of settling time on DAC0800.

EQUIPMENTS:-

- BEL-ADA Board
- Power supply
- Function generator
- DSO.

THEORY:-

The DAC0800 series are monolithic 8-bit high-speed current-output digital-to-analog converters (DAC) featuring typical settling time of 100ns. when used as a multiplying DAC monotonic performance over a 40 to 1 reference current range is possible. The DAC0800 are also features high compliance complementary current output to allow differential output voltage of 20V p-p with simple resistor load. The noise immune inputs of the DAC0800 series will accept TTL level with the logic threshold pin, V_{LC} , grounded. Changing the V_{LC} potential will allow direct interface to other logic families. The performance and characteristics of the device are essentially unchanged over the full ± 4.5 to ± 18 V power supply range; power dissipation is only 33mW with ± 5 V supplies and is independent of the logic input states.

FEATURES OF DAC0800

- Full scale error: ± 1 LSB
- Nonlinearity over temperature: $\pm 0.1\%$
- Full scale current drift: ± 10 ppm/ $^{\circ}\text{C}$
- High output compliance: -10 V to 18V

- Complementary current output
- Interface directly with TTL, CMOS, PMOS and others
- 2 quadrant wide range multiplying capability
- Wide power supply range: $\pm 4.5\text{V}$ to $\pm 18\text{V}$
- Low power consumption: 33mW at $\pm 5\text{V}$
- Low cost
- Fast Settling output Current: 100ns

SETTLING TIME

The settling time of a DAC is the time from a change of digital code to when the output comes within and remains within some error band as shown in Figure below with the error band will almost invariably be ± 1 or $\pm 1/2$ LSB.

The settling time of a DAC is made up of four different periods: the switching time or dead time (during which the digital switching, but not the output, is changing), the slewing time (during which the rate of change of output is limited by the slew rate of the DAC output), the recovery time (when the DAC is recovering from its fast slew and may overshoot), and the linear settling time (when the DAC output approaches its final value in an exponential or near-exponential manner). If the slew time is short compared to the other three (as is usually the case with current output DACs), the settling time will be largely independent of the output step size. On the other hand, if the slew time is a significant part of the total, the larger the step, the longer the settling time.

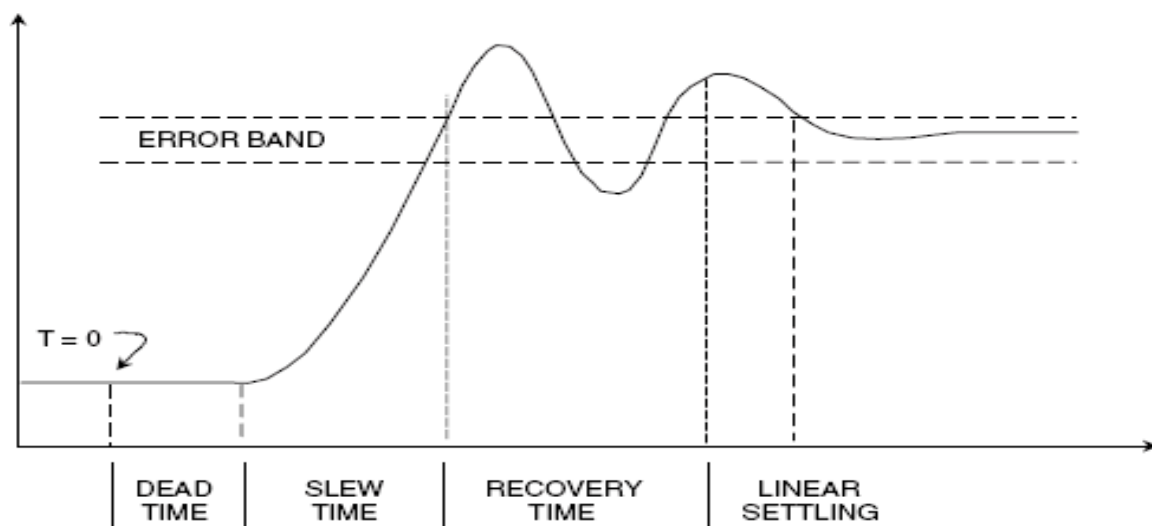


FIGURE-3.1

EFFECT OF SETTLING TIME

The importance of settling time in a data conversion system is that certain analog operations must be performed in sequence, and one operation may be accurately settled before the next operation can be initiated. In DACs this parameter gives information about

the time required by the converter to meet the right output value after a change in the input code.

Less settling time results in fast conversion and consumes less time than more settling time.

PIN DESCRIPTION

Pin 1: Threshold control

Pin 2: I_{OUT}

Pin3: V^-

Pin4: I_{OUT}

Pin5 to Pin 12-Digital Input (MSB to LSB)

Pin13: V^+

Pin14: $V_{REF}(+)$

Pin 15: $V_{REF}(-)$

Pin 16: Compensation

PROCEDURE:-

UNIPOLAR OPERATION

1. Make the connection as shown below in FIGURE-3.2.
2. Apply the digital input signals from the onboard Input Switches (SW1-SW8) to the 8 BIT MONOLITHIC DAC and connect the jumper J3 in unipolar mode.
3. Observe the output at VOUT of 8 BIT MONOLITHIC DAC using DSO for various digital input data pattern.

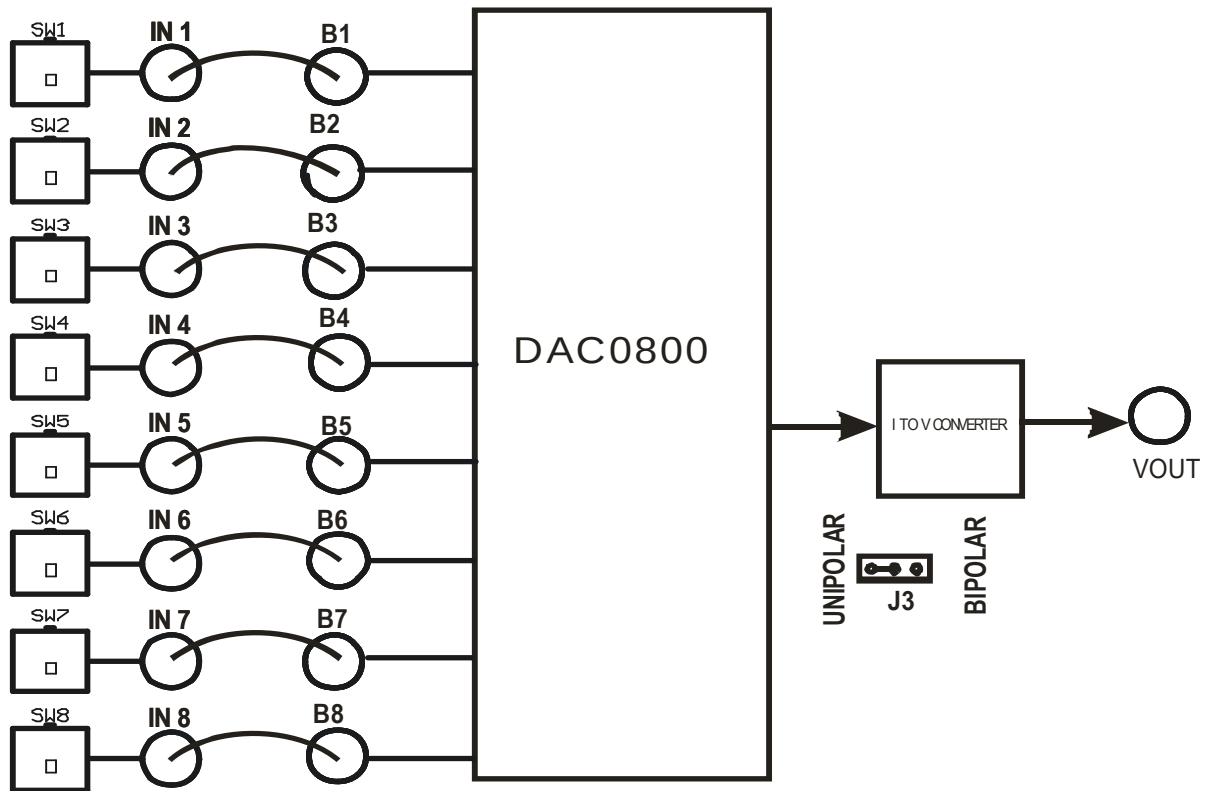


FIGURE-3.2

OBSERVATIONS:-**UNIPOLAR MODE**

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Observed O/P Voltage(V)
0	0	0	0	0	0	0	0	00.004
0	0	0	0	0	0	0	1	0.040
0	0	0	0	0	0	1	1	0.114
0	0	0	0	0	1	1	1	0.262
0	0	0	0	1	1	1	1	0.545
0	0	0	1	1	1	1	1	1.13
0	0	1	1	1	1	1	1	2.32
0	1	1	1	1	1	1	1	4.01
1	0	0	0	0	0	0	0	4.59
1	1	0	0	0	0	0	0	6.96
1	1	1	0	0	0	0	0	8.14
1	1	1	1	1	0	0	0	9.05
1	1	1	1	1	1	0	0	9.21
1	1	1	1	1	1	1	0	9.29
1	1	1	1	1	1	1	1	9.33

RAMP IN UNIPOLAR MODE

PROCEDURE:-

- 1) For Observing Ramp output, connect the inputs(B1-B8) of DAC0800 8 BIT MONOLITHIC DAC section with the onboard BINARY COUNTER(Q0-Q7) respectively having clock frequency nearby 100KHz (CLK IN) as shown below in FIGURE-3.3.
- 2) Connect the jumper J3 in unipolar mode and observe the ramp signal at VOUT of 8 BIT MONOLITHIC DAC in DSO.

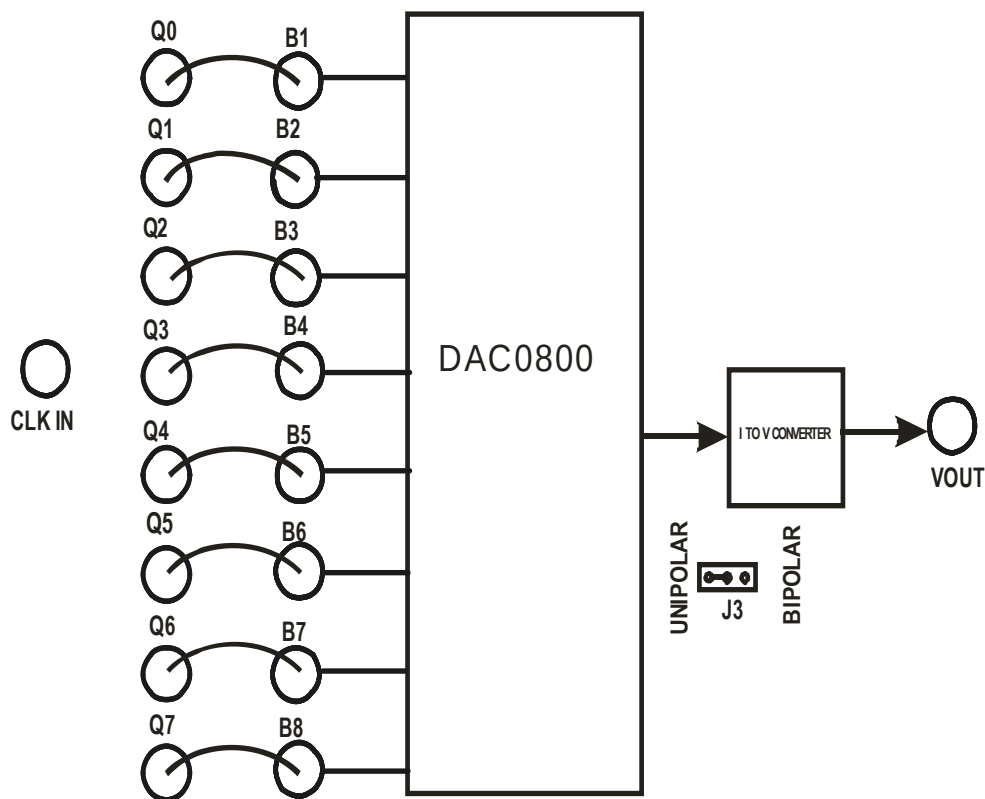


FIGURE-3.3

- 3) If waveform shown in FIGURE-3.4 is not coming then adjust the POT PR5 (offset adjust) until you will get a ramp of nearly 9.5V.

CAPTURED WAVEFORM:-

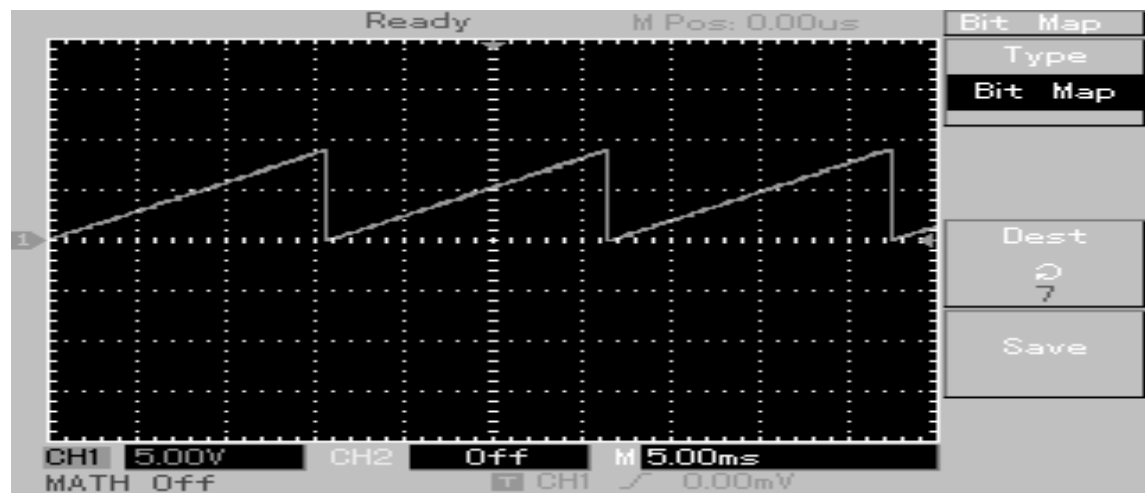


FIGURE-3.4

BIPOLAR OPERATION

PROCEDURE:-

- 1) Connect the inputs(B1-B8) of 8 BIT MONOLITHIC DAC with the onboard input switches(SW1-SW8) and connect jumper J3 in bipolar mode as shown in FIGURE-3.5
- 2) Connect the jumper J3 in bipolar mode and observe the ramp signal at VOUT of 8 BIT MONOLITHIC DAC in DSO.

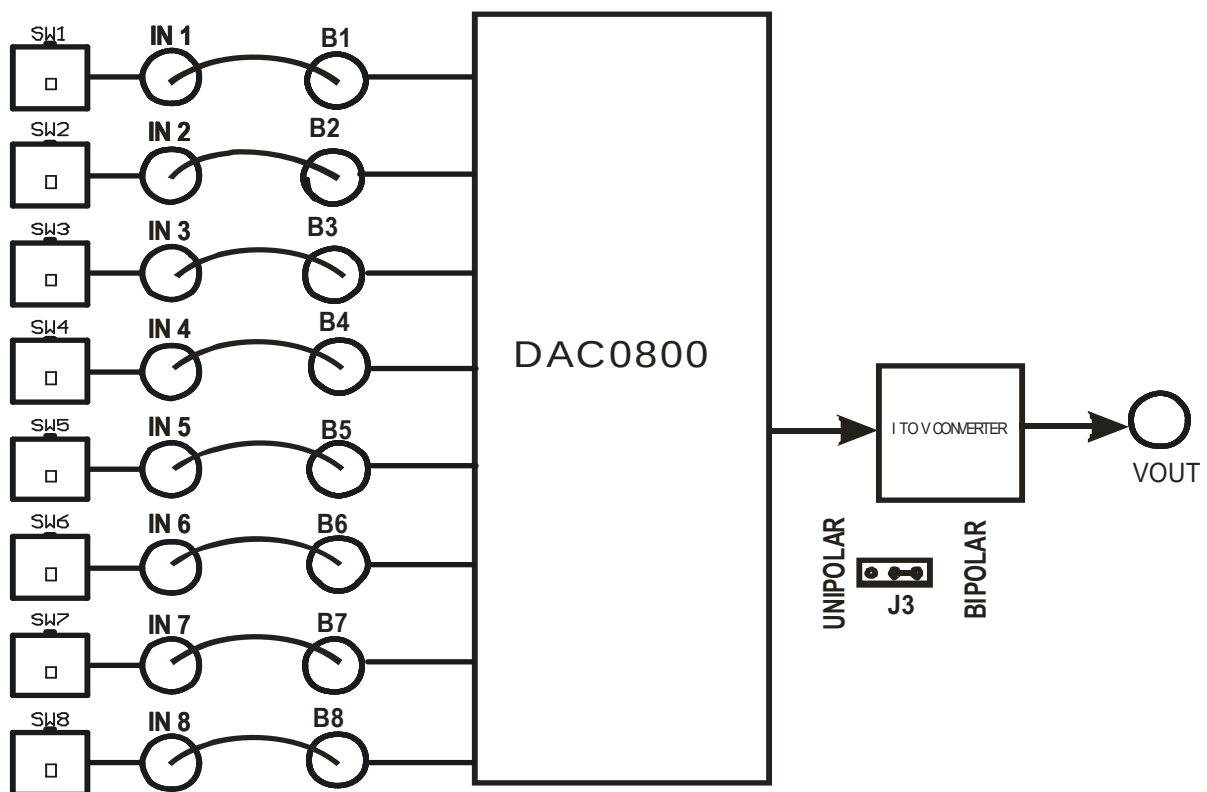


FIGURE-3.5

OBSERVATIONS:-**BIPOLAR MODE**

2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0	Observed O/P Voltage
0	0	0	0	0	0	0	0	-8.93V
0	0	0	0	0	0	0	1	-8.92V
0	0	0	0	0	0	1	1	-8.90V
0	0	0	0	0	1	1	1	-8.84V
0	0	0	0	1	1	1	1	-8.25V
0	0	0	1	1	1	1	1	-7.08V
0	0	1	1	1	1	1	1	-4.73V
0	1	1	1	1	1	1	1	-40.8mV
1	0	0	0	0	0	0	0	47mV
1	1	0	0	0	0	0	0	4.61V
1	1	1	0	0	0	0	0	7.09V
1	1	1	1	0	0	0	0	8.26V
1	1	1	1	1	0	0	0	8.85V
1	1	1	1	1	1	0	0	9.14V
1	1	1	1	1	1	1	0	9.29V
1	1	1	1	1	1	1	1	9.36V

RAMP IN BIPOLAR MODE**PROCEDURE:-**

- 1) For Observing Ramp output, connect the inputs(B1-B8) of DAC0800 8 BIT MONOLITHIC DAC section with the onboard BINARY COUNTER (Q0-Q7) respectively having clock frequency nearby 100KHz(CLK IN) as shown below in FIGURE-3.6.
- 2) Connect the jumper J3 in bipolar mode and observe the ramp signal at VOUT of 8 BIT MONOLITHIC DAC in DSO.

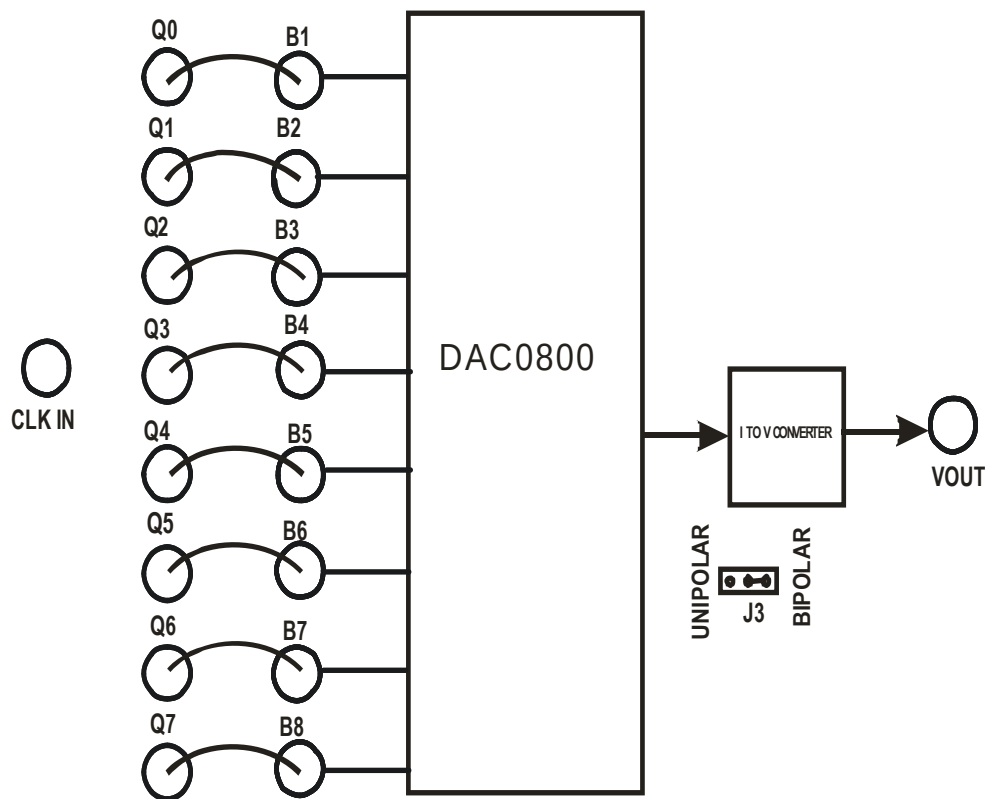


FIGURE-3.6

- 3) If the waveform shown in FIGURE-3.7 is not coming then adjust the POT PR5 (offset adjust) until you will get a ramp of nearly $\pm 9.5\text{Vpp}$.

CAPTURED WAVEFORM:-

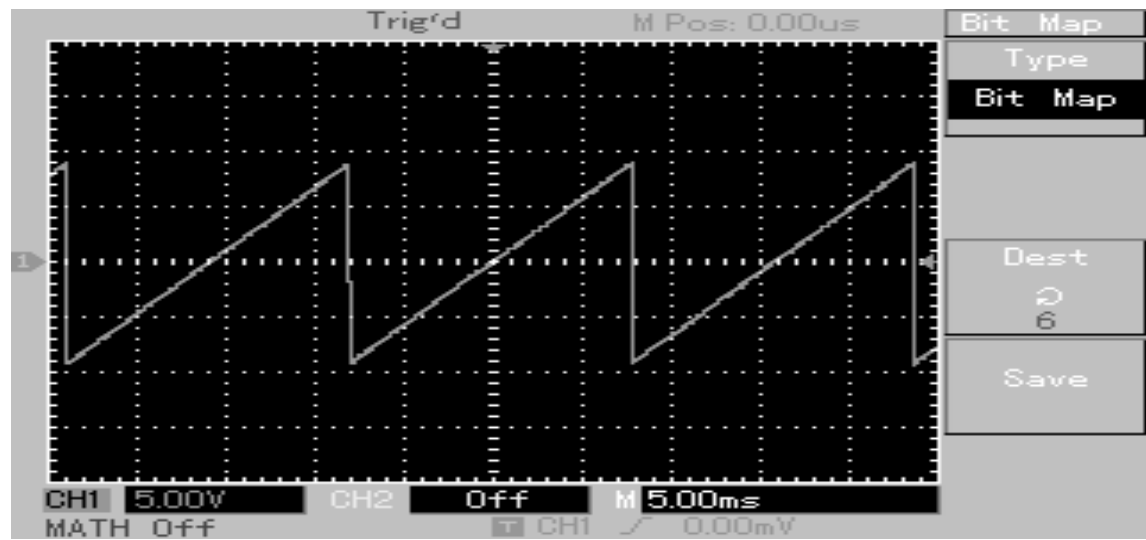


FIGURE-3.7

SETTLING TIME MEASUREMENT

PROCEDURE:-

1. To measure the settling time, give TTL square wave of 1 KHz from Function generator to all the input (B1-B8) of the 8 BIT MONOLITHIC DAC by shorting all the input together as shown in FIGURE-3.8
2. Observe the transition at VOUT of the 8 BIT MONOLITHIC DAC and measure the time required to change from output high to output low state. This will give the fall time of DAC0800 shown in FIGURE-3.11
3. The time required to change from O/P low to O/P high will give the Rise time of DAC 0800. Measure the Rise time as shown in FIGURE-3.10
4. These two readings will give the settling time of whole system i.e. DAC 0800 and LM741.

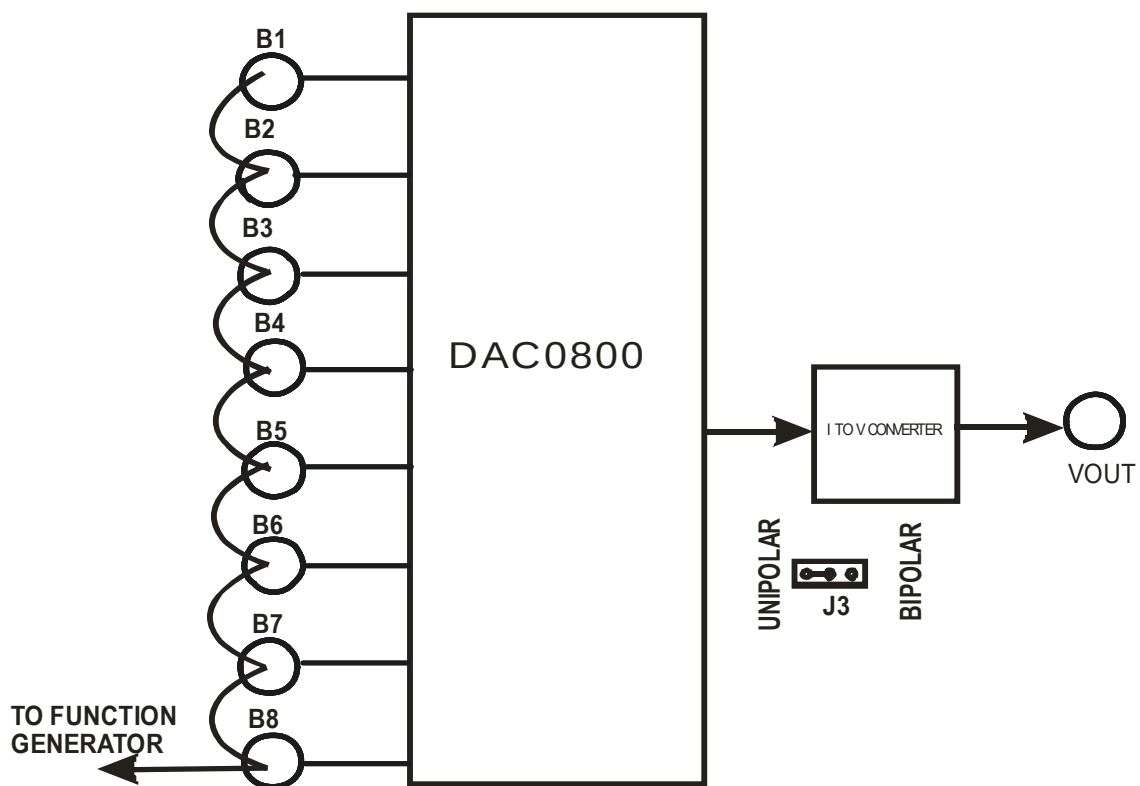


FIGURE-3.8

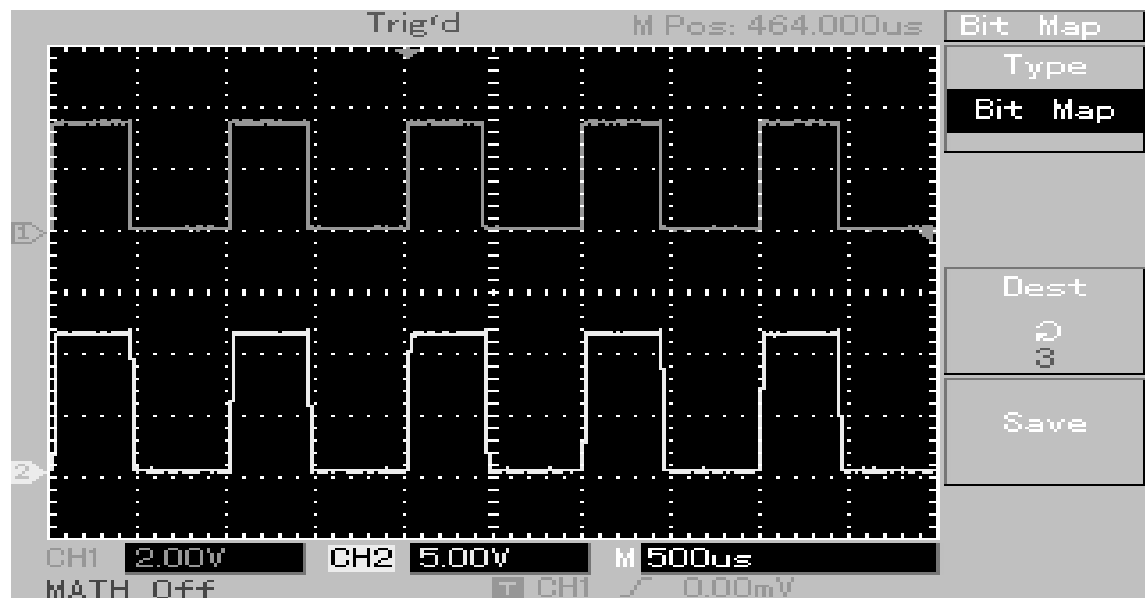
CAPTURED WAVEFORM:-**INPUT AND OUTPUT WAVEFORMS OF DAC0800**

FIGURE-3.9

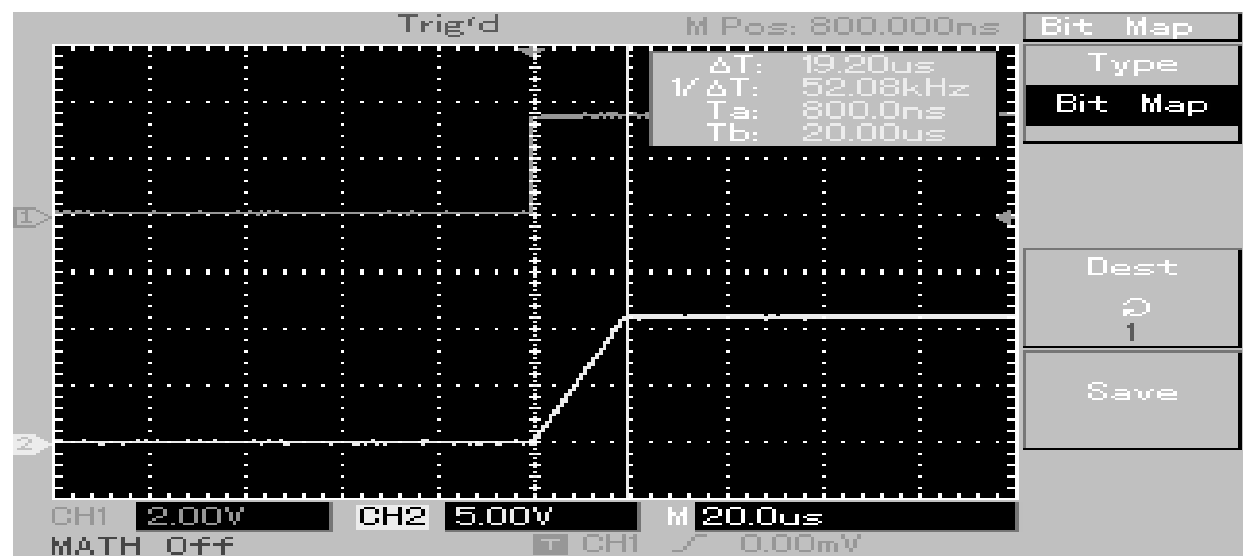
RISE TIME MEASUREMENT

FIGURE-3.10

FALL TIME MEASUREMENT

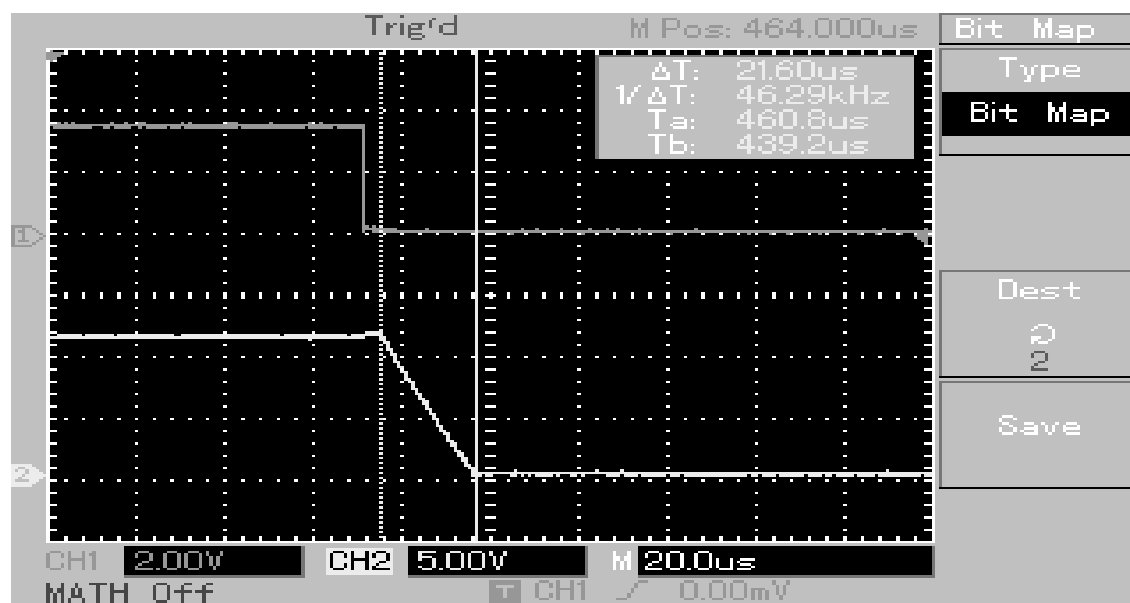


FIGURE-3.11

Readings from DSO

Rise Time (Micro Sec.)	Fall Time (Micro Sec.)
19.20	21.60

Effects of Settling Time

More settling time results in deformations in waveforms that is transitions are less sloppy. We can see this by increasing the frequency to 10 K Hz as shown in FIGURE-3.12

Frequency -10KHz

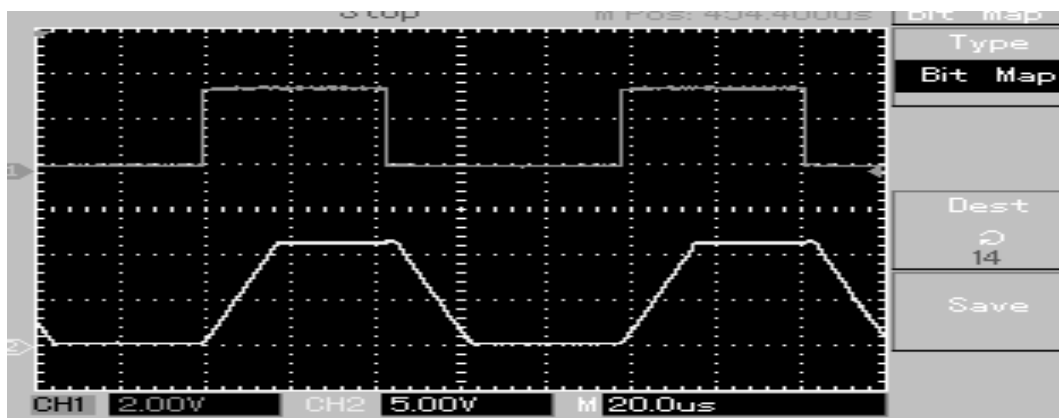


FIGURE-3.12

Frequency -54K Hz.

At this frequency ON-TIME of input signal is equal to the RISE-TIME of output waveform. Hence we are getting a triangular waveform. Since fall time of output is less than off time of input we are getting some flatness in lower edge of output signal as shown in FIGURE-3.13

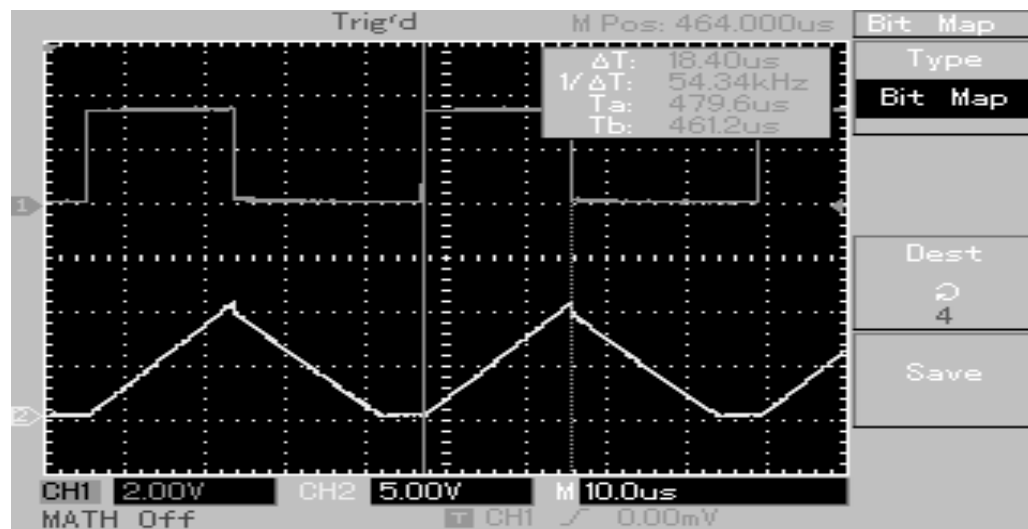


FIGURE-3.13

CONCLUSION:-

DAC0800 is used for digital to analog conversion in unipolar as well as bipolar mode and output voltage depends on the applied digital input voltage and should not exceed the applied reference voltage. Settling time can be observed from the DSO at different input frequency.

EXPERIMENT NO.4

EXPERIMENT NO.4

NAME:-

Working of A/D conversion using counter ramp method

OBJECTIVE:-

To study 8 bit Analog to Digital Conversion using counter ramp method for various digital data pattern.

EQUIPMENTS:-

- BEL-ADA Board
- Power supply,
- Function generator
- DSO.

THEORY:-

The process of changing an analog signal to an equivalent digital signal is accomplished by the use of an A/D converter. It is also referred to as an encoding device since it is used to encode signals for entry into a digital system.

It is of following types.

- Counter ramp converter
- Successive approximation converter
- Dual slope converter
- Parallel converter

Now in this experiment we will see the working of counter ramp type converter in details

It is also known as the stair step-ramp, or simply counter A/D converter. The basic idea is to connect the output of a free-running binary counter to the input of a DAC, then compare the analog output of the DAC with the analog input signal to be digitized..

The counter is continuously clocked, and the up/down control line is driven by the output of the comparator. So, when the analog input signal exceeds the DAC output, the counter goes into the “count up” mode. When the DAC output exceeds the analog input, the counter switches into the “countdown” mode. Either way, the DAC output always counts in the proper direction to *track* the input signal.

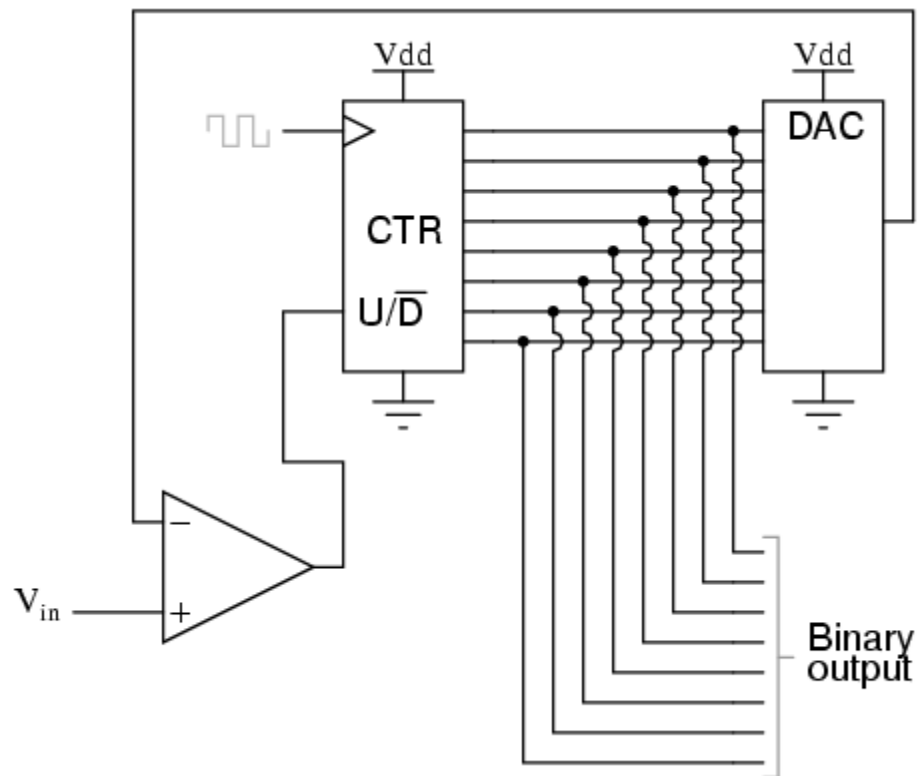
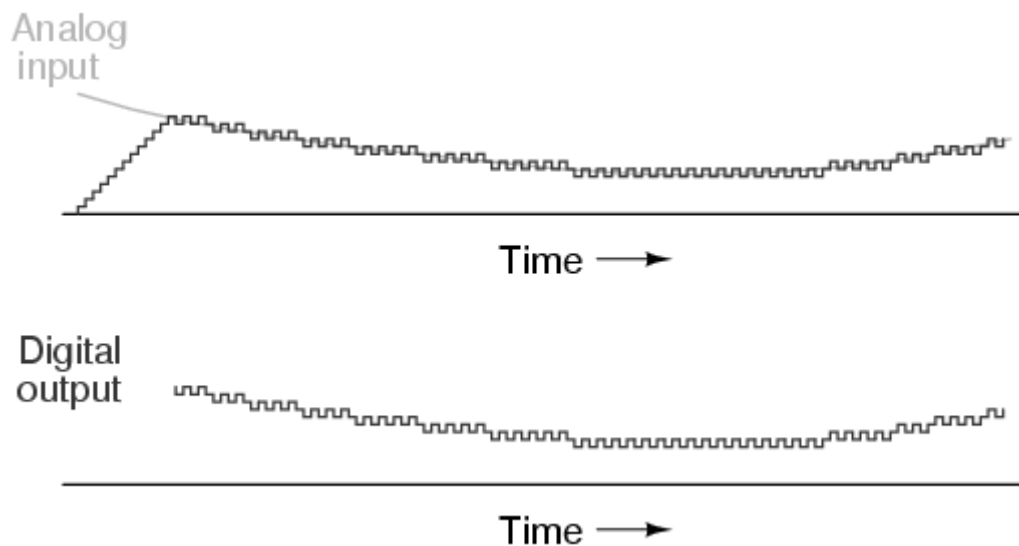


FIGURE-4.1

An advantage of this converter circuit is speed, since the counter never has to reset. Note the behavior of this circuit:



It has much faster update time than any of the other “counting” ADC circuits. Perhaps the greatest drawback to this ADC design is the fact that the binary output is never stable: it always switches between counts with every clock pulse, even with a perfectly stable analog input signal. This phenomenon is informally known as *bit bobble*, and it can be problematic in some digital systems.

This tendency can be overcome, though, through the creative use of a shift register. For example, the counter's output may be latched through a parallel-in/parallel-out shift register only when the output changes by two or more steps. Building a circuit to detect two or more successive counts in the same direction takes a little ingenuity, but is worth the effort

PROCEDURE:-

1. Make the connections as shown in the FIGURE-4.2
2. Give the TTL level clock signal of frequency near about 100KHz to the CLK IN of COUNTER RAMP ADC
3. Give the onboard VARIABLE DC input (0V-12V) signal to the ANALOG IN of comparator that you want to be digitized. Note that this section should be used only in unipolar configuration i.e. only 0V-12V input signal should be applied at ANALOG IN.
4. Now connect the output of the COUNTER RAMP ADC to the OUTPUT LEDs provided on the board and observe the analog equivalent digital output.
5. At 0V input, 2 LSB will continue glowing due to the offset error of the whole circuit. Near 10.56V all LEDs will glow.
6. Till 11V DAC OUT will continuously track the Variable DC input voltage and after that it goes in oscillations.
7. Apply onboard 1KHz sine wave and see the DAC output on DSO, it will appear to be triangular because it will continuously track the analog input and counter will count up and down as the input voltage increases or decreases with respect to reference voltage i.e. DAC OUT.
8. Load switch SW25 is used to reset the system.

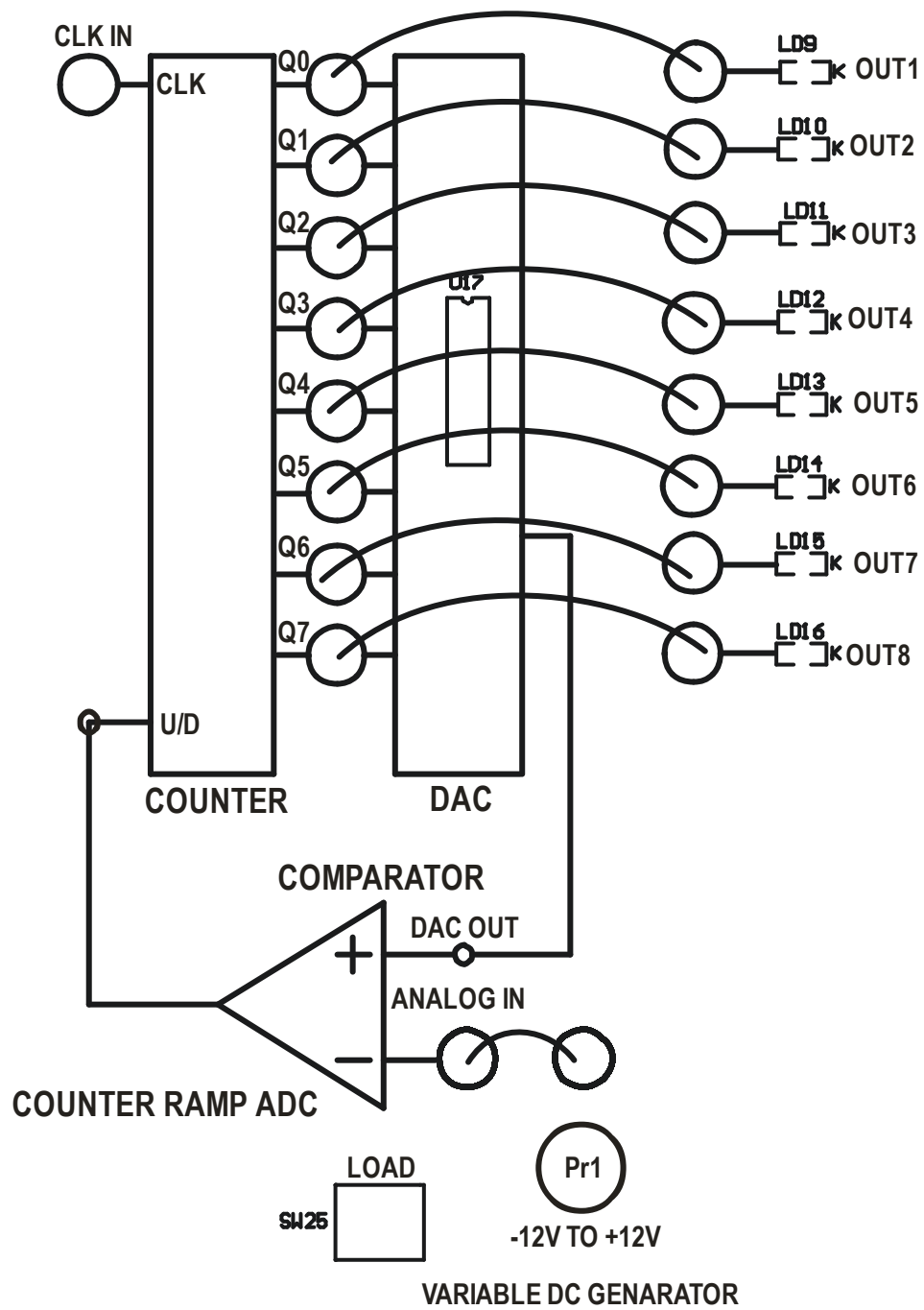


FIGURE-4.2

CONCLUSION:-

Counter ramp ADC is used to track the analog input voltage to the corresponding digital output voltage .

EXPERIMENT NO.5

EXPERIMENT NO.5

NAME:-

A/D conversion using monolithic IC LTC1278 to do full scale adjustment for unipolar and bipolar signals.

OBJECTIVE:-

To study working of 12 bit Analog to Digital Conversion using monolithic IC LTC1278 and do full scale adjustment for unipolar and bipolar signals.

EQUIPMENTS:-

- BEL-ADA Board
- Power supply
- DSO

THEORY:-

LTC 1278 is a 1.6 μ S, 500ksps, sampling 12-bit A/D converter that draws only 75mW from a single 5V or $\pm$ 5V supplies. Unipolar and Bipolar conversion modes add to the flexibility of the ADC .It converts 0V-5V unipolar inputs from a single 5V supply and $\pm$ 2.5V bipolar inputs from $\pm$ 5V supplies. The internal clock is trimmed for 1.6 μ S conversion time .The clock automatically synchronizes to each sample command, eliminating problems with asynchronous clock noise found in other devices. A separate convert start input and a data ready signal(/busy) ease connections to FIFOs, DSPs and microprocessors.

PROCEDURE:-

FULL SCALE ADJUSTMENT FOR UNIPOLAR SIGNAL

1. For zero full-scale error apply an analog input of 4.99817V (i.e., FS – 1 1/2LSB or last code transition) at the input and adjust PR4 until the LTC1278 output code flickers between 11111111 1110 and 1111 1111 1111.

UNIPOLAR CONFIGURATION

1. Make the connection as shown inFIGURE-5.1. Press the slider switch SW26 in Unipolar mode and remove the jumper J4.
2. Apply onboard variable DC input voltage (0V-5V) to the 12 BIT ADC and measure the equivalent digital output voltage by connecting the output pins(D0-D11) to the onboard output LEDs.
3. Apply 0V to the Input and you will observe that no LED will glow and in 5V all LEDs will glow.

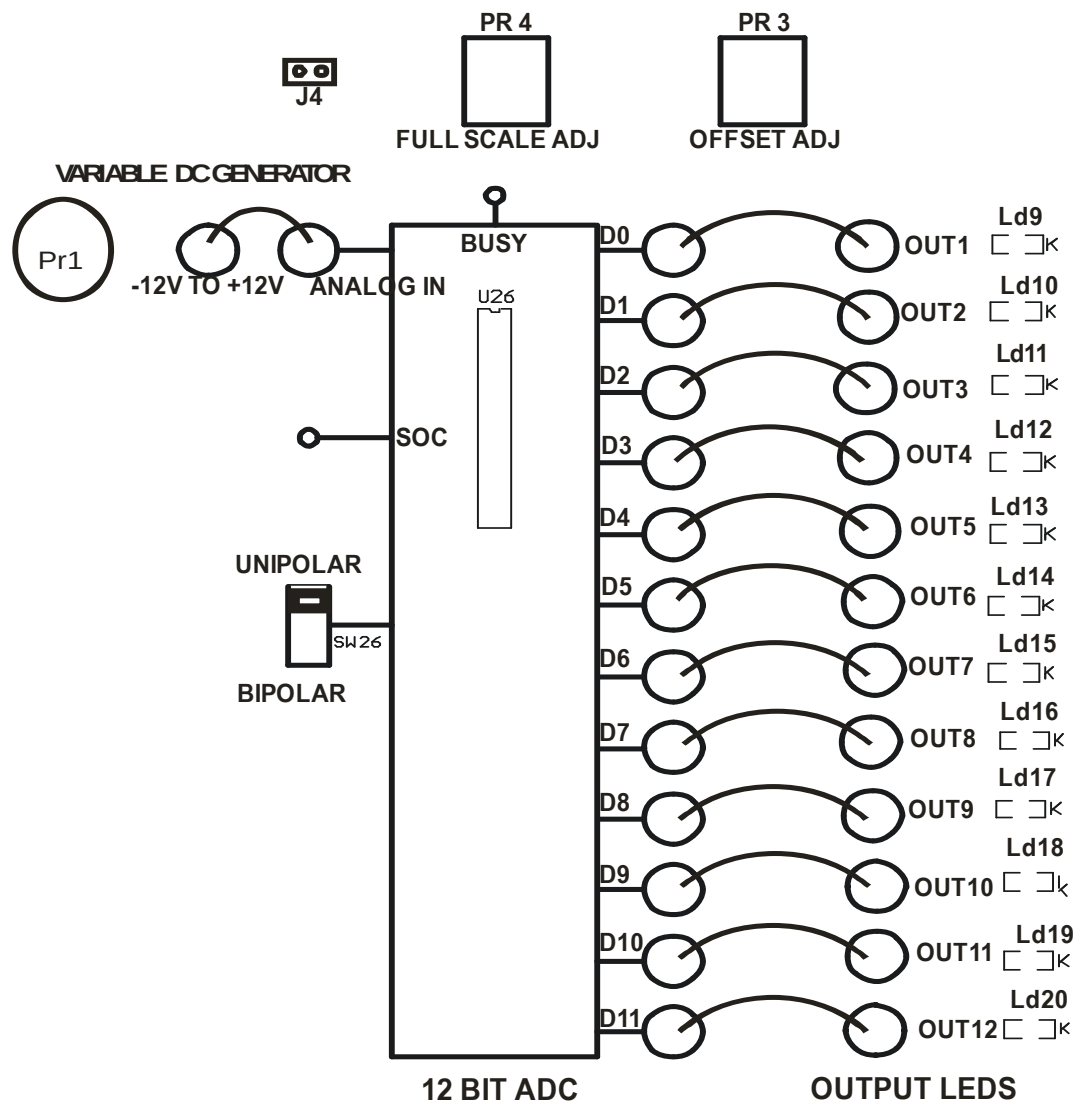


FIGURE-5.1

FULL SCALE ADJUSTMENT FOR BIPOLAR SIGNAL

1. Bipolar offset error adjustment is achieved by trimming PR4. This is done by applying an input voltage of 2.49817V (FS – 1.5LSBs) and PR4 is adjusted until the output code flickers between 0111 1111 1110 and 0111 1111 1111.

PROCEDURE:-

1. Make the connection as shown in FIGURE-5.1
2. Make the 12 BIT ADC in Bipolar configuration by moving the slider switch SW26 and connect the jumper J4.
3. Give onboard variable DC input voltage ($\pm 2.5V$) to the IC and measure the equivalent digital output voltage by connecting the output pins to the onboard OUTPUT LEDs.
4. Apply 0V to the ANALOG IN of 12 BIT ADC, output code will flicker between 0000 0000 and 1111 1111 1111.
5. Now increase the voltage up to +2.5V, output code will flicker between 0111 1111 1110 and 0111 1111 1111.
6. Now decrease the voltage up to -2.5V, output code will be 1000 0000 0000.
7. Verify the results on output leds.

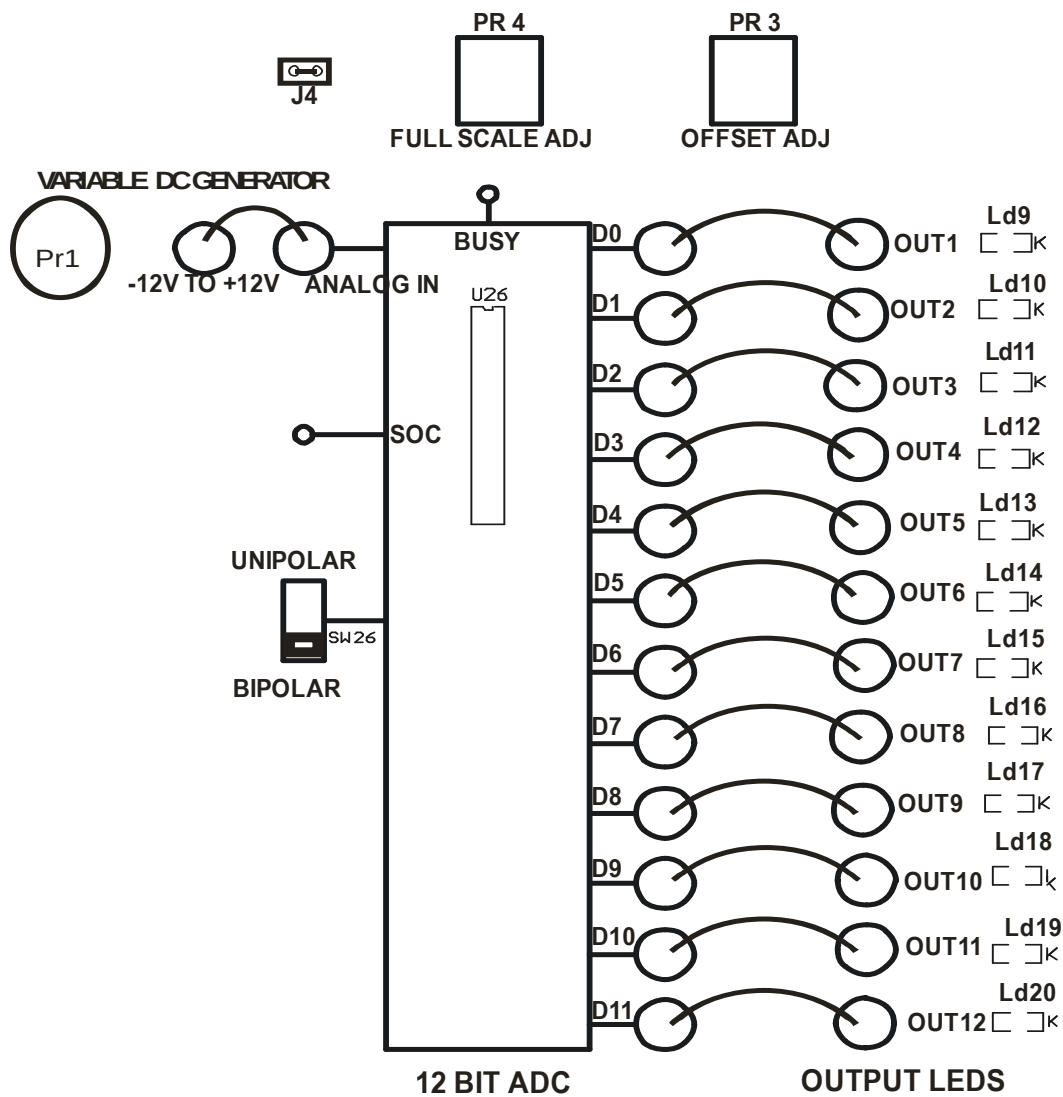


FIGURE-5.2

CONCLUSION:-

LTC1278 is a 12 bit Analog to Digital Converter IC which converts 0V-5V unipolar inputs from a single 5V supply and $\pm 2.5V$ bipolar inputs from $\pm 5V$ supply.

EXPERIMENT NO.6

EXPERIMENT NO.6

NAME:-

To measure the conversion time of ADC

OBJECTIVE:-

Study to measure the conversion time of ADC

EQUIPMENTS:-

BEL-ADA board, its power supply and DSO

THEORY:-

CONVERSION TIME

The time between the start and end of the conversion is known as the conversion time. The positive pulse on CONVST pin starts the conversion and when BUSY pin is high, it indicates that conversion is completed. We can measure the time between the two using DSO. Given conversion time of ADC: 1.6uS

PROCEDURE:-

1. Make the connection as shown in FIGURE-6.1.
2. Give onboard analog input signal of 1 KHz to the ANALOG IN of 12 BIT ADC and then connect the probes of DSO to the IC pins where we have the SOC and BUSY test points.
3. Now observe the waveform on DSO, we can measure the conversion time using cursor of DSO.
4. Bring one cursor at the falling edge of SOC and other cursor at the rising edge of BUSY pulse as shown in FIGURE-6.2
5. The time between these two is the total conversion time.

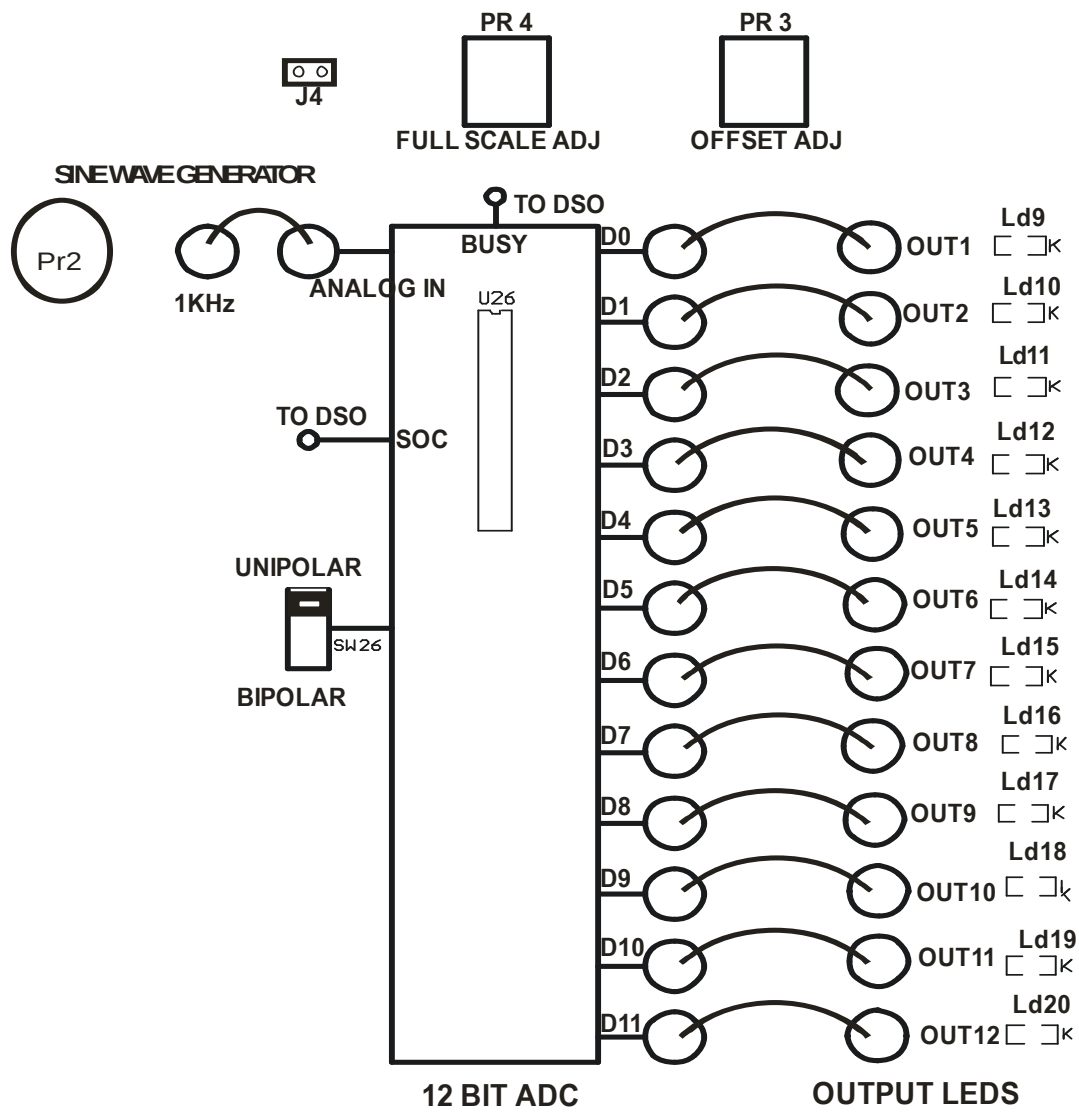


FIGURE-6.1

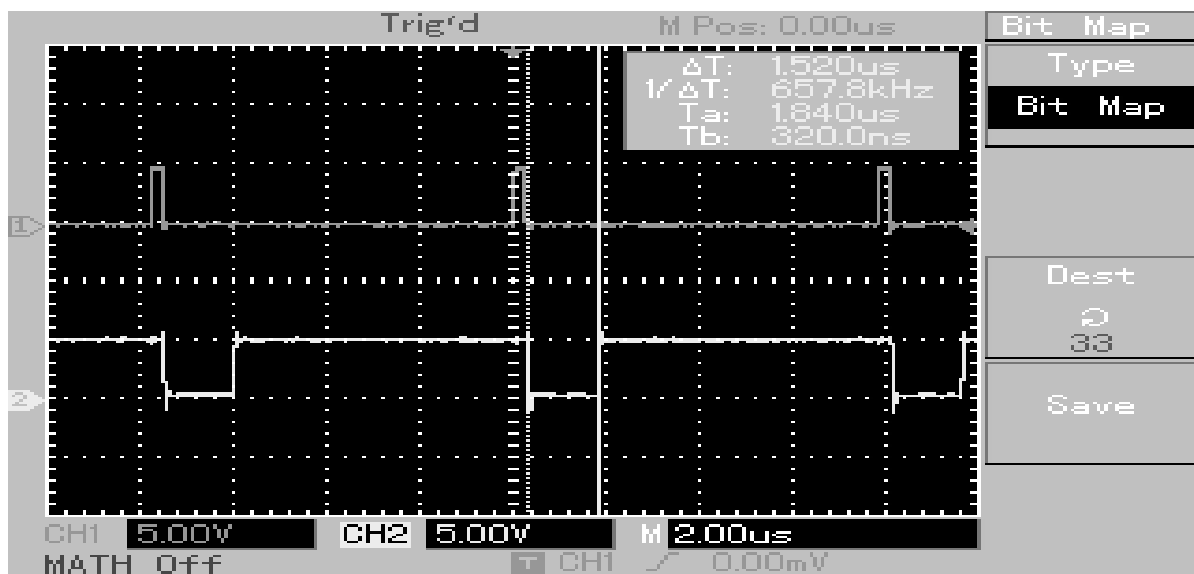
CAPTURED WAVEFORM:-

FIGURE-6.2

CONVERSION TIME OBSERVED: 1.520us**CONCLUSION:-**

LTC1278 is 500ksps, sampling 12-bit A/D converter which has conversion time of 1.6uS. By observing the values of give conversion time of IC and measured conversion time of IC we can conclude that both are nearly same.

EXPERIMENT NO.7

EXPERIMENT NO.7

NAME:-

To measure the resolution of ADC

OBJECTIVE:-

Study to measure the resolution of ADC

EQUIPMENTS:-

- BEL-ADA Board
- Power supply
- DSO

THEORY:-

The resolution of a n-bit analog-to-digital Converter (ADC) is a function of how many parts the maximum signal can be divided into. The formula to calculate resolution is 2^n . For example, a 12 bit ADC has a resolution of $2^{12} = 4,096$. Therefore, our best resolution is 1 part out of 4,096, or 0.012% of the full scale.

An ADC takes an analog signal and turns it into a binary number. Thus, each binary number from the ADC represents a certain voltage level. Resolution is the smallest input voltage change a digitizer can capture. Resolution can be expressed in bits (LSB), in proportions, or in percent of full scale.

Resolution limits the precision of a measurement. The higher the resolution (number of bits), the more precise the measurement. An 8-bit ADC divides the vertical range of the input amplifier into 256 discrete levels. With a vertical range of 5 V, the 8-bit ADC cannot ideally resolve voltage differences smaller than 19 mV. In comparison, a 12-bit ADC with 4096 discrete levels can ideally resolve voltage differences as small as 1.2mV

RESOLUTION OF UNIPOLAR ADC

For calculating resolution of 12 bits unipolar input analog voltage we have following formula:

$$\text{Resolution} = 1/2^{12} \times V_{\text{ref}}$$

Where V_{ref} is the applied unipolar input voltage range and 12 is the no. of bits of ADC.

PROCEDURE:-

1. Make the connection as shown in FIGURE-7.1 in 12 BIT ADC Section, move the slider switch SW26 in unipolar mode, remove the jumper J4.

2. Give on board variable DC input voltage to the ANALOG IN of 12 BIT ADC and observe the equivalent digital output voltage by connecting the output pins to the onboard OUTPUT LEDs.
3. Take the difference of applied voltage when all LEDs are glowing (V_{max}) and when all LEDs are off (V_{min}).
4. This will give the voltage range of ADC.

$$V_{ref} = (V_{max} - V_{min})$$

$$V_{min}(\text{observed}) = 0V$$

$$V_{max}(\text{observed}) = 4.99V$$

$$V_{ref} = 4.99V$$

$$\text{Resolution} = 1/2^{12} \times V_{ref} = 4.99/4096 = 0.0012V$$

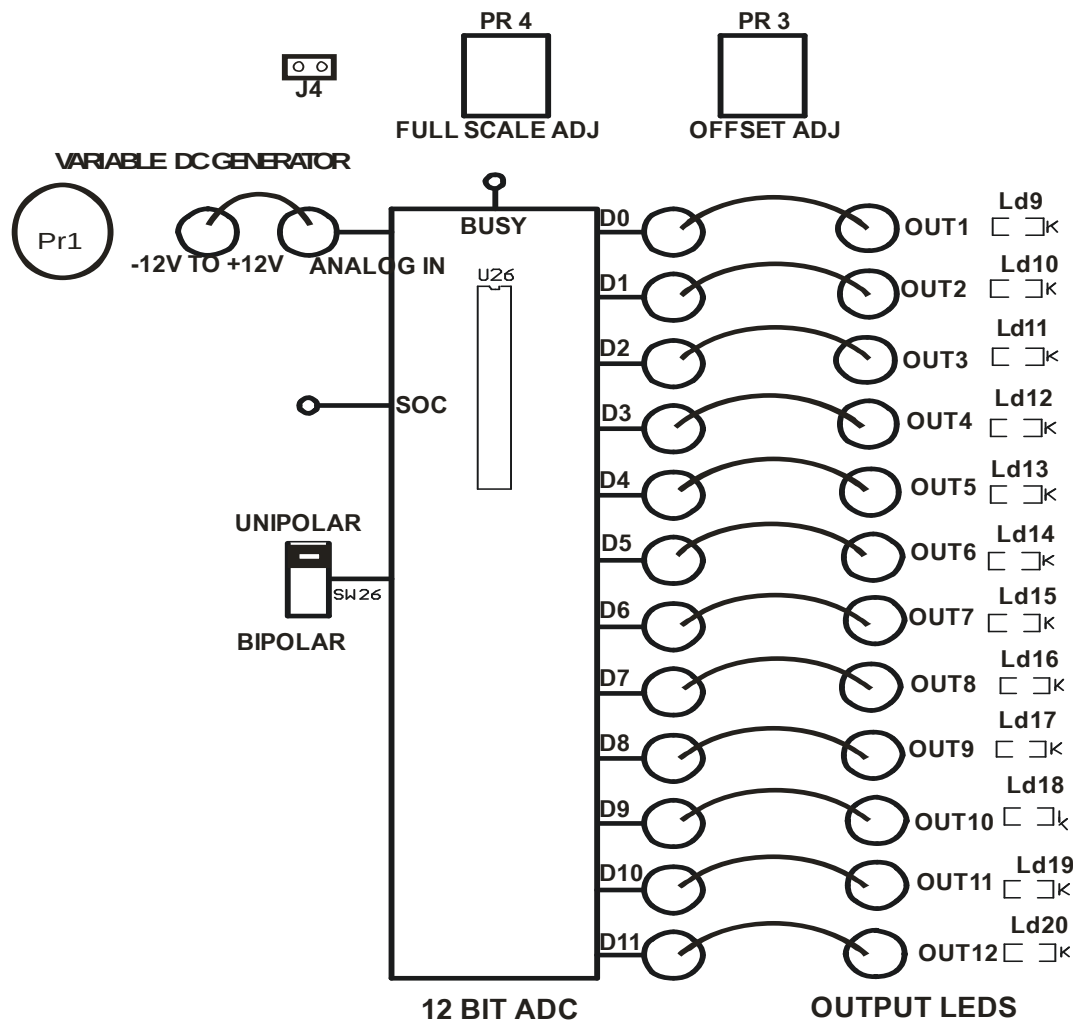


FIGURE 7.1

RESOLUTION OF BIPOLAR ADC

For calculating the resolution of 12 bits bipolar input ADC we have following formula

$$\text{Resolution} = 1/2^{12-1} \times V(\text{ref})$$

Where $V(\text{ref})$ is the difference between applied maximum and minimum bipolar input voltage to the ADC and 12-1 i.e. 11 is the number of bits of ADC and the 12th bit (MSB) indicates the sign of input voltage. 0 indicates the positive input voltage and 1 indicates the negative input voltage.

PROCEDURE:-

1. Make the connection as shown in FIGURE-7.2, move the slider switch SW26 in bipolar mode, connect the jumper J4.
2. Give on board variable DC input voltage to the ANALOG IN of 12 BIT ADC and observe the equivalent digital output voltage by connecting the output pins to the onboard OUTPUT LEDs.
3. Observe the glowing of LEDs representing the bipolar analog equivalent digital output. Take the difference of applied voltage when all LEDs are glowing (V_{max}) except MSB and when only MSB is glowing (V_{min}). This is $V_{(ref)}$.
 $V_{min} \text{ observed} = -2.49V$
 $V_{max} \text{ observed} = +2.49V$
 $V_{(ref)} = (V_{max} - V_{min}) = 4.98$

$$\text{Resolution} = 1/2^{12-1} \times V_{(ref)} = 4.98/4096 = 0.0012V$$

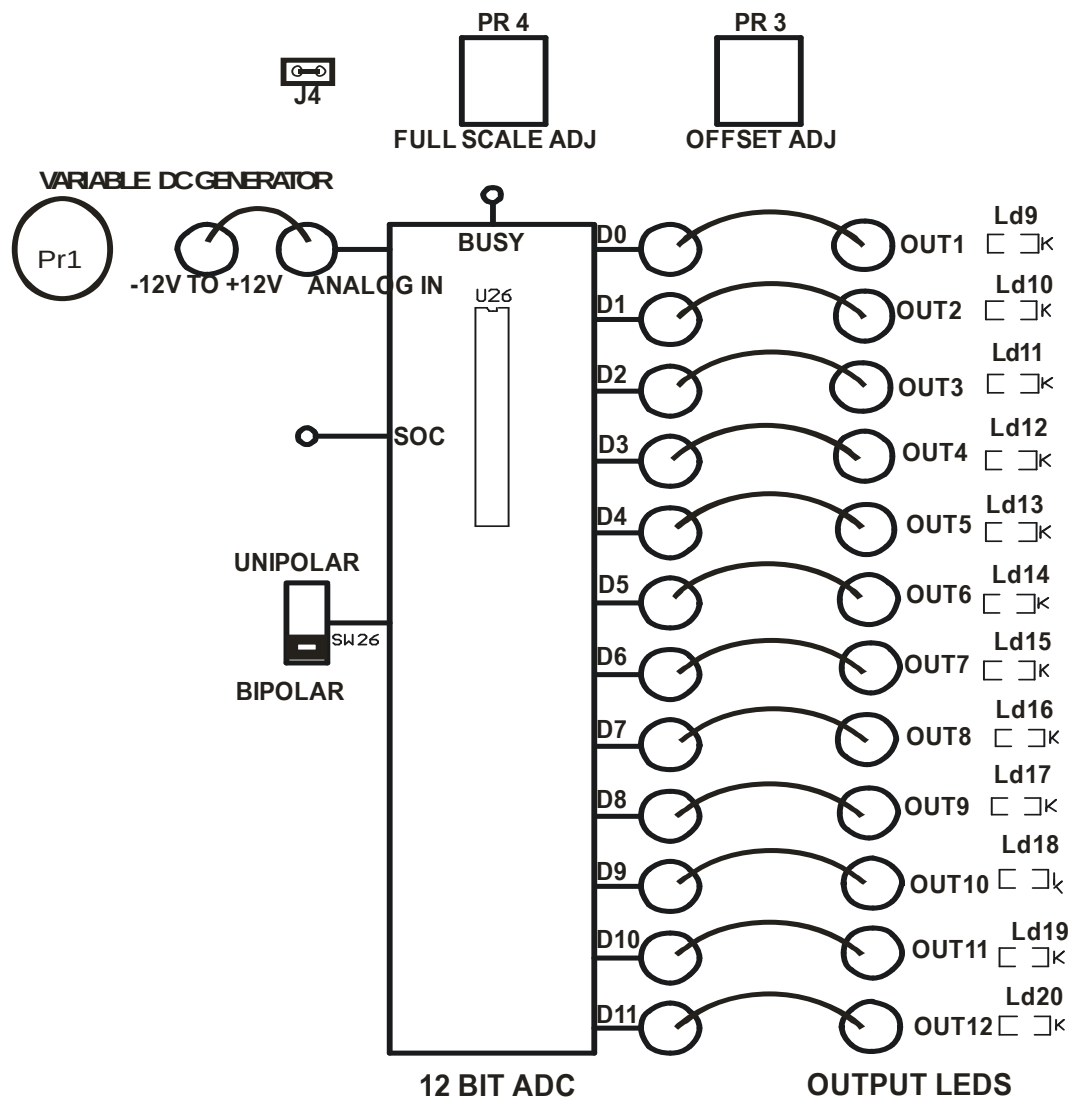


FIGURE-7.2

CONCLUSION:-

LTC1278 is a 12-bit ADC with 4096 discrete levels that can ideally resolve voltage differences as small as 1.2mV. 12 bit ADC has more resolution than 8 bit ADC with the same reference voltage.